



Recent advances in CO₂ capture and mineralization using layered double hydroxide-based materials: A review

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Abstract

The continuous release of substantial amounts of carbon dioxide (CO₂) to the atmosphere has resulted in numerous severe adverse effects. Several materials have been synthesized and utilized for CO₂ capture. One class of such materials is layered double hydroxides (LDHs), which have emerged as promising materials for CO₂ capture due to their tunable properties, high surface area, and excellent CO₂ adsorption capabilities. Although there are some review articles on CO₂ capture and conversion using various materials, there is still a notable lack of thorough reviews focusing on the utilization of LDH-based materials for CO₂ capture. Additionally, the field of CO₂ capture and mineralization using LDH-based materials is rapidly evolving, necessitating up-to-date comprehensive reviews to analyze, evaluate, and condense the dispersed information found in recently published research articles. Accordingly, this review article provides a comprehensive overview of recent advancements in CO₂ capture using LDH-based materials. After briefly introducing the topic, different synthesis protocols of LDH-based materials are briefly reviewed. Then, CO₂ capture using LDHs, calcined LDHs, impregnated LDHs, composites containing LDHs, amine functionalized LDHs, and during steam methane reforming, are thoroughly analyzed and discussed. Additionally, the effects of synthesis method and post treatment of LDH-based materials on CO₂ capture, effect of modification and functionalization on LDHs, and the effects of various process conditions including temperature, pressure, water vapor, and gas composition on the performance of CO₂ capture by LDH-based materials are reviewed. Limitations, challenges, obstacles, and remaining knowledge gaps are highlighted, and future research works to address them are proposed.

Keywords Carbon dioxide (CO₂) capture · Layered double hydroxides (LDHs) · LDH composites · Impregnated LDHs · Adsorption mechanism · Regeneration

Introduction

CO₂ is a greenhouse gas that has a serious impact on global warming and climate change. In the past few decades, there has been a significant increase in the concentration of CO₂ in the atmosphere due to human activity, mostly from the burning of fossil fuels (Ramirez-Corredores et al. 2023; Nunes

2023; Zhumadilova et al. 2023). Environmental studies project a continuous increase in the atmospheric concentration of CO₂ due to the dependence on fossil fuels as the main source of energy, coupled with an expected increase of 53% in energy consumption by 2030 (Ramirez-Corredores et al. 2023). Carbon capture, utilization, and storage (CCUS) technologies play a crucial role in the execution of decarbonization endeavors, aiming primarily to address climate change by compensating for a substantial variation of 100 to 1000 gigatons (Gt) of CO₂ emissions throughout the twenty-first century. The failure to implement effective CCUS technologies may lead to a significant increase in mitigation costs, potentially ranging from 29 to 297%, in its pursuit of decreasing atmospheric CO₂ levels to 450 ppm by the end of the century. Thus, carbon capture is experiencing significant expansion in the fields of sustainable chemistry and separation sciences. The recovered CO₂ from various industrial sources, such as cement, steel, and refineries,

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could be used as a feedstock for various useful products. Furthermore, there have been suggestions to use recovered CO₂ as a primary resource for innovative applications, such as artificial photosynthesis and photocatalysis. This methodology has resulted in advancements in several domains, including process-enhanced efficiency, computational modeling, reactor configuration, and materials science (Zou and Rodrigues 2001; Roger 2023).

Direct air capture (DAC) is a powerful approach to combat climate change by actively decreasing the amount of CO₂ in the atmosphere (Abdelnaby et al. 2024). The effectiveness of DAC systems is heavily dependent on the development of efficient and scalable sorbent materials (Tao et al. 2023). Research efforts on the use of solid adsorbents for CO₂ capture have increased in response to the problem of rising greenhouse gas levels (Querejeta et al. 2023; Rashid and Rafeq 2023). Metal–organic frameworks (MOFs) (Al-Rowaili et al. 2021), zeolites (Hanh et al. 2024), amine-functionalized compounds such as amine-functionalized magnesium oxides (Yong et al. 2000; Alkadhém et al. 2020, 2021), and LDHs (Ge et al. 2023) are promising materials for selective CO₂ capture. Mixed oxides derived from LDHs have shown a significant promise as efficient options for CO₂ capture, particularly within the temperature range of 200 to 400 °C (Rashid and Rafeq 2023). Adsorbents based on LDHs have several advantages, including reduced energy requirements for regeneration and enhanced durability throughout multiple cycles. LDHs are highly sought after for their rapid adsorption kinetics, as well as their excellent performance in water-based pre-combustion CO₂ capture and equilibrium-related CO₂ processes (Ai et al. 2023).

LDHs alternatively are referred to as hydrotalcite-like compounds or anionic clays, exhibiting a geometry characterized by a series of stacked layers resembling brucite. These layers are interspersed with anions and water molecules, which possess positive charges and serve to counterbalance the charges. The chemical composition of LDH minerals may be expressed by the general formula: $[M^{2+}_{1-x}M^{3+}_x(OH)_2]^{x+}[A^{n-}]_{x/n}mH_2O$, where M²⁺ typically refers to a metal ion with a charge of +2, such as Ca²⁺, Mg²⁺, Zn²⁺, Ni²⁺, Co²⁺, Mn²⁺, or Fe²⁺, while M³⁺ refers to a metal ion with a charge of +3, such as Al³⁺, Cr³⁺, Mn³⁺, Fe³⁺, Co³⁺, or Ni³⁺. A^{n−} represents an anion with a variable charge, denoted as *n*, such as CO₃^{2−}, NO₃[−], PO₄^{3−}, SO₄^{2−}, or Cl[−]. The ratio *x* of M²⁺/M³⁺ is typically between 0.1 and 0.5 (Cornu et al. 2022; Tang et al. 2022; Bahadi et al. 2024a, b, c).

The adaptability of LDHs is attributed to their highly modifiable characteristics, rendering them well-suited for a wide range of applications, including catalysis, fire retardancy, polymer-LDH nanocomposites, CO₂ capture, luminescence, and magnetism. LDHs exhibit a notable inclination towards the adsorption of acidic CO₂ (Sharma et al. 2023). Thermal treatment facilitates the affinity by

stimulating dehydroxylation and decarbonation activities through the release of interlayer water. Consequently, the creation of a three-dimensional network demonstrates a larger surface area and improved resilience when exposed to high temperatures (Betz et al. 2016).

As the research on CO₂ capture and mineralization using LDH-based materials is rapidly evolving, there is a growing need for current and thorough reviews to consolidate, analyze, and evaluate the dispersed information found in recently published research articles. Such comprehensive and up-to-date reviews would provide researchers and professionals with easier and faster tracking of the advancements in the field of CO₂ capture using LDH-based materials. Thus, this article presents a comprehensive and contemporary analysis and discussion of the latest literature on CO₂ capture using LDH-based materials. While there exist numerous review articles on CO₂ capture, the focus of this review article is on the utilization of LDH-based materials for CO₂ capture and mineralization. Recent and comprehensive reviews on CO₂ capture using LDHs-based materials are still relatively lacking in the published literature. Different key aspects of LDHs utilization in CO₂ capture will be reviewed in this article. Additionally, the effects of LDHs chemical composition, synthesis methods, and modification strategies on the enhancement of CO₂ capture will be explored. Furthermore, the mechanisms of CO₂ capture by the LDHs-based materials and the regeneration of the LDH-based material will be discussed. The remaining key gaps, challenges, and obstacles will be highlighted, and future research work to address them will be proposed at the end of this review.

Synthesis of LDHs

LDHs are a kind of structured material that consists of several layers of cationic metals. The metals under consideration exhibit either divalent (M²⁺) or trivalent (M³⁺) characteristics. The interstitial regions between these layers strategically place hydrated anions to maintain charge equilibrium. When synthesizing LDHs, several factors must be carefully considered. These factors include the choice of metallic cations, temperature, duration of aging, pH level, and method of assembly. There are many methods available for synthesizing LDHs. The techniques included in this category include salt oxidation, reconstruction, ion exchange, coprecipitation, urea hydrolysis, the sol–gel method, hydrothermal treatment, and microwave-assisted synthesis. Depending on the specific requirements of the LDHs being produced, any of these methods may be used; each has its advantages (Klemkaite et al. 2011; Naidu et al. 2020; Jiang et al. 2023). The manufacture of LDHs from waste materials has emerged as a significant topic of interest because of its considerable potential for extensive industrial applications.

Method of co-precipitation

The creation of LDH hybrids mostly relies on the co-precipitation method. The experimental procedure entails the dissolution of an inorganic salt in an alkaline solution while maintaining a consistent pH level in order to regulate the molecular size and structure of the hybrids. The presence of an interlayer anion in salt often results in the mixing of solutions containing metal ions and hydroxyl anions inside the layers. A 0.1 N NaOH titration is used to sustain a pH level above 8 and expedite the process of co-precipitation. The final product is obtained by subjecting the precipitate to a minimum aging period of 24 h at room temperature, followed by centrifugation, washing with deionized water, and vacuum drying. The production of biomolecule-LDH hybrids involves the combination of biomolecules with exchangeable interlayer anions present in LDH particles (Barahuie et al. 2014).

The manipulation of temperature and aging duration enables the regulation of particle size, with shorter aging periods leading to the formation of smaller particles (Li et al. 2013a). To produce LDHs, Miyata altered critical factors, including reactant concentration, pH, and washing conditions (Zahedi et al. 2022). Before introducing the selected interlayer anion into water, it is necessary to progressively introduce solutions containing the appropriate molar ratio of divalent and trivalent cations. To facilitate the co-precipitation of the two metal salts, an alkaline solution is introduced into the pH throughout the operation. A research done by Amer et al. (2021) showed the efficacy of using 3-amino-1H-1, 2, 4-triazole to modify a ternary Mg-Cu-Al LDH (MLDH). The research findings revealed that the MLDH and LDH produced exhibited high efficacy as adsorbents for the removal of cadmium and copper ionic species. The assessment of the adsorption process included the consideration of elements such as the mass of the adsorbents, the impact of temperature, cycle duration, concentration of ionic species in the medium, and the acidity of the medium (Amer et al. 2021). The co-precipitation method, as shown in Fig. 1, illustrates a crucial technique used in LDH synthesis.

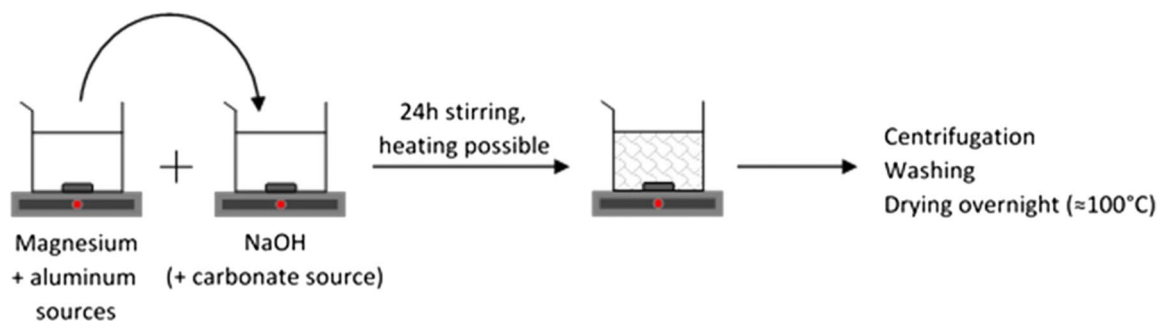
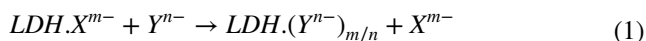


Fig. 1 The co-precipitation process is used to synthesize LDH (Chaillot et al. 2021)

The ion exchange method

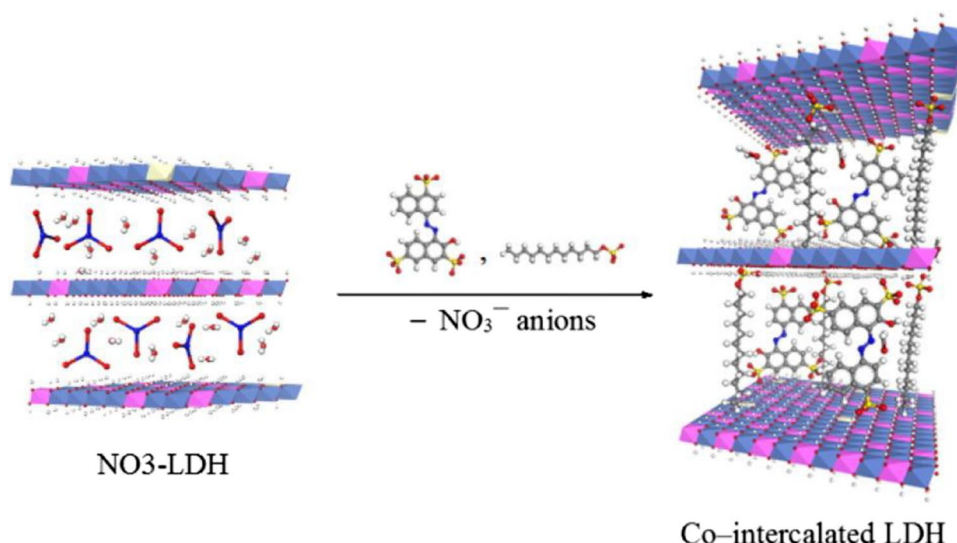
The ion exchange technique is often considered an indirect strategy when co-precipitation or urea procedures fail to provide the necessary interlayer anions (Santamaría et al. 2023). The process begins with the formation of LDH by co-precipitation with NO_3^- , CO_3^{2-} , and Cl^- since host anions with single charges exhibit greater ease of exchange compared to those with multiple charges. The requisite anions are then introduced into the interlayer region by the agitation of the LDH precursor inside a solution that accommodates a greater quantity of the desired anion for intercalation. In order to prevent carbonate intercalation, it is necessary to use an inert environment during the anion exchange process (Mishra et al. 2018; Jijoe et al. 2021). Figure 2 illustrates the ion-exchange process where nitrate anions (NO_3^-) are exchanged with larger organic molecules to form co-intercalated layered double hydroxide (LDH) structures. On the left, the NO_3 -LDH structure shows nitrate anions within the interlayer spaces of the LDH. During the ion-exchange process, the nitrate anions are replaced by co-intercalated organic molecules, as depicted on the right, resulting in a modified LDH with enhanced properties, suitable for specific applications such as catalysis or adsorption (Sanati and Rezvani 2018):



Method of reconstruction (memory effect)

This approach involves subjecting metal salts to a consistent heating rate of 5 °C/min for 4 h at a temperature of 500 °C in a nitrogen environment. The generated material is combined with a guest molecule in a solution of distilled water, maintaining a regulated pH level. The process involves allowing the precipitate to reach maturity at ambient temperature, followed by filtration, washing

Fig. 2 Visual representation of the ion exchange process (Sanati and Rezvani 2018)



with deionized water, and subsequent drying using a vacuum (Mascolo and Mascolo 2015). The incorporation of carbon microspheres has been seen in the calcination-reconstruction process. The nanohybrids consisting of methotrexate (MTX) intercalated with LDH (MTX/LDH) exhibit an irregular structure in the absence of a carbon microsphere template. However, the use of a monodispersed template composed of carbon spheres leads to the formation of a regular structure. Figure 3 illustrates the memory effect of LDH regeneration using a graphic depiction.

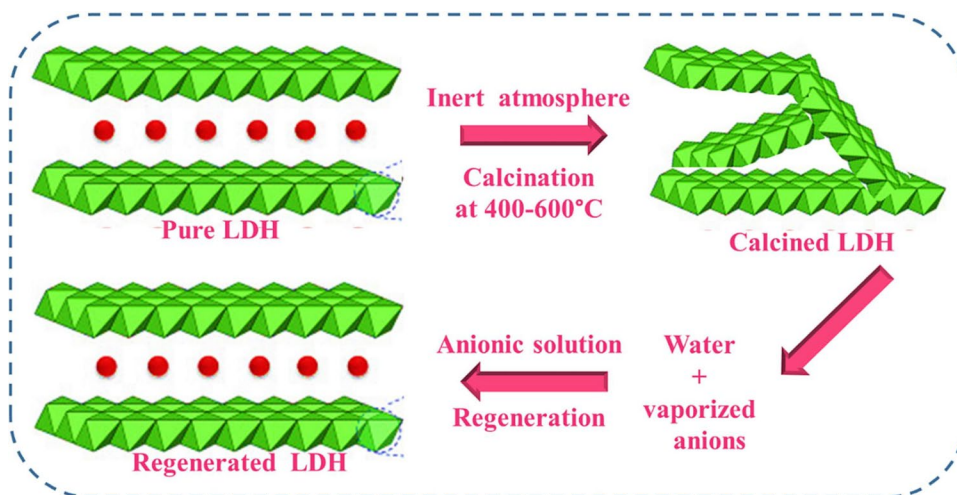
The method of urea hydrolysis

Urea, being very soluble in water, acts as a gentle Bronsted base and may be used as a precipitating agent to increase pH levels via heat-induced breakdown. Commencing at 90 °C, the gradual breakdown of urea results in a pH elevation

of up to 10 and a reduction in the degree of supersaturation. Urea has a crucial role in regulating pH levels and promoting the creation of monodispersed LDH materials characterized by exceptional crystallinity and purity. During hydrolysis, CO₃^{2–} anions are generated and function as interlayer anions (Verzì et al. 2020). The next passage delineates the prevailing amalgamations of coprecipitation and hydrothermal synthesis, including the process of urea hydrolysis. According to the findings of Berber et al. (2013), the optimal approach for achieving consistent particle production throughout synthesis included the manipulation of urea concentration, the aging process, and M²⁺/M³⁺ molar ratios. Additional studies have been carried out using coprecipitation and urea hydrolysis in conjunction with the calcination of Ni–Al–Mg hydrotalcite (Berber et al. 2013).

In their study, Inayat et al. (2011) investigated the formation of Zn–Al-LDHs using urea hydrolysis in a uniform environment. Nitrate was used as the anion to regulate the

Fig. 3 Illustration depicting the memory effect in the context of LDH regeneration (Farhan et al. 2024)



charge in the interlayer gap. It has been shown that by using a combination of these methods, each with varying periods of synthesis, it is possible to distinguish the borders that determine whether nitrate or carbonate will act as the anion between layers. A notable hindrance to the widespread adoption of this approach is the enduring presence of carbonate, which arises from the breakdown of urea (Inayat et al. 2011).

To address this issue, the introduction of anions, such as chloride, into the carbonate particles inside the LDH structure was achieved through the use of mineral acid or salt and acid treatment (Cao et al. 2022). The disadvantages of these therapies include a higher time commitment and increased effort required for post-treatment strategies. Modifying the treatment conditions improperly might result in a decrease in the size of crystallites, the degree of crystallinity, and the loss of substance owing to the partial degradation of LDH in the acidic environment (Schlack et al. 2021). Costantino et al. (1998) proposed an alternate method for converting carbonate-containing LDHs into the chloride structure after blending. To prevent LDH disintegration, the LDHs underwent a gentle HCl gas treatment at a temperature of 150 °C. The process of urea hydrolysis produces thin plates characterized by a narrow particle size distribution and a shorter aging period in comparison to coprecipitation. This is attributed to the slow reaction rate and little presence of contaminants in urea hydrolysis. Possible adverse outcomes of urea hydrolysis synthesis include breakdown and the creation of carbonates as counter-anions (Costantino et al. 1998). Figure 4 illustrates the process of synthesizing hydroxalcalcites through urea hydrolysis.

The sol–gel approach

The sol–gel method is well recognized for its ability to effectively produce nanoscale particles that exhibit excellent uniformity, purity, and a specific surface area, all while maintaining a low cost. By manipulating the amount of reactants and the time of aging through the introduction or elimination of reactant species, it is possible to modify the structural characteristics of the final product using this methodology. The experimental protocol involves the dissolution

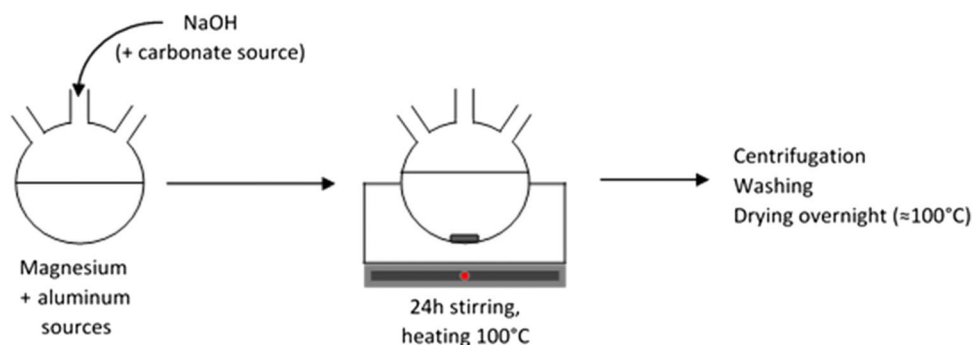
of suitable metal salts in water under ambient conditions (Lee et al. 2016).

Wei et al. (2022a) presented a flexible sol–gel method for producing LDHs that may be tailored to include certain metal cations. The morphology exhibited extensive pore size ranges ranging from 3 to 5 nm, together with substantial specific surface areas reaching up to 290 m²/g. Nanocapsular morphology is used to replace platelet-like particles produced by co-precipitation and other techniques (Wei et al. 2022a). When rehydrated, the sol–gel method shows that it can calcine and rehydrate artificial hydroxalcalcite while keeping its lamellar structure. This finding demonstrates the potential use of this method in medical applications (Ishikawa et al. 2020; Waqar et al. 2024). The sol–gel technique is widely favored in many applications owing to its user-friendly characteristics and the superior performance of the materials it generates. There is no text provided. Nevertheless, using this synthetic approach leads to a decrease in crystallinity, necessitating the use of supplementary techniques like as microwave irradiation, hydrothermal treatment, and ultrasonication. The sol–gel process provides benefits in the manufacturing of tiny particles that have a significant specific surface area and outstanding cleanliness. Figure 5 illustrates the hydroxalcalcite formation through the sol–gel synthesis technique.

The hydrothermal method

Hydrothermal treatment under moderate circumstances has been used to promote the synthesis of LDHs, hence improving the crystallinity and particle size (Liu et al. 2023). The temperature might gradually increase to 150 °C over a period of many days. The metallic salts are dissolved in water and subjected to a temperature range of 30 to 300 °C, while maintaining steam pressure, during various stages of synthesis. Afterward, the solution is transferred to a stainless steel reactor under hydrothermal conditions. Lin et al. (2019) conducted a study where they created Cu–Zn–Al hydroxalcalcite catalysts to examine how increasing the temperature from 35 to 140 °C affects the process of reducing arsine. Data interpretation becomes challenging due to the

Fig. 4 Illustration depicting the process of synthesizing hydroxalcalcites by the hydrolysis of urea (Chaillot et al. 2021)



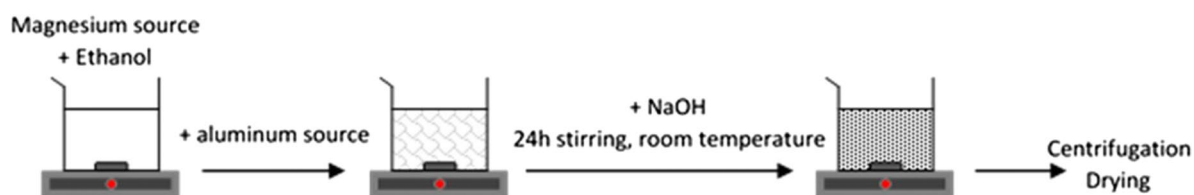


Fig. 5 Synthesis of hydrotalcites via sol–gel process (Chaillot et al. 2021)

existence of other parameters such as temperature, molar ratio, and calcination time (Jang et al. 2014; Lin et al. 2019).

The principal reactants often used in various applications are hydroxides and metal oxides. Ogawa and Asai (2000) used gibbsite and brucite as initial materials to investigate the structural characteristics of natural minerals (Ogawa and Asai 2000). Hydrothermal treatment is reported to improve crystallinity, purity, and particle size; however, it is a time-consuming and energy-intensive process. Labuschagne et al. (Labuschagné et al. 2018) showed a sustainable method for creating a structure similar to hydrotalcite by studying the process of hydrothermal precipitation using untreated magnesium oxides and aluminum hydroxide (Labuschagné et al. 2018). Figure 6 illustrates the hydrothermal synthesis technique used for the manufacture of hydrotalcite.

Solvothermal method

The solvothermal synthesis of carbonate-type layered double hydroxide (LDH) monolayer nanosheets is significantly influenced by solvent selection based on characteristic parameter matching (Wang et al. 2024). This approach enhances the formation and stability of these nanosheets, which are crucial for various applications. This method allows precise control over the size, shape, and composition of nanostructures, making it ideal for applications like energy storage (Khapate et al. 2023).

This method involves dissolving metal precursors in a solvent, typically ethanol, and using an alkali source like urea to initiate the formation of hydroxides. The mixture is heated in a sealed, high-pressure environment, such as a Teflon-lined autoclave, at temperatures ranging from 120 to 200 °C. The study by Yu et al. emphasizes the importance of matching the surface free energy (γ) and solubility

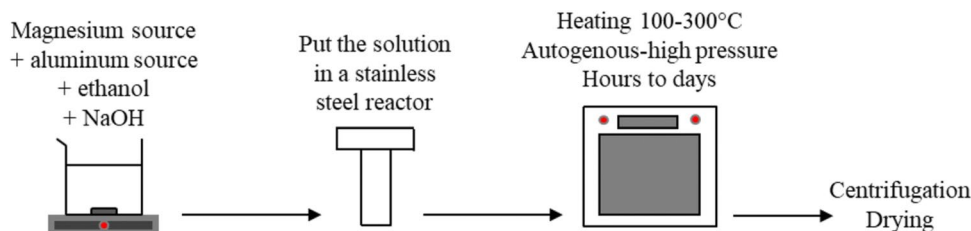
(δ) parameters of solvents with those of the LDHs. This matching is critical for achieving monolayer thickness (~ 1 nm) in the synthesized nanosheets. Utilizing a combination of solvents can optimize the synthesis process, leading to stable dispersions of LDHs for extended periods (up to 6 months) (Meng et al. 2017).

For instance, in synthesizing cobalt/nickel LDHs, cobalt and nickel nitrates are dissolved in ethanol, with urea added as an alkali source. The solution is heated at 120 °C in a Teflon-lined autoclave for 3 h. During heating, urea decomposes, forming ions that react with the metal ions to create LDH nanoparticles, which then grow into two-dimensional nanosheets that self-assemble into three-dimensional hierarchical structures, often exhibiting flower-like morphologies. This unique morphology is influenced by the properties of ethanol (Wang et al. 2023b). After cooling, the product is filtered, washed, and dried, resulting in mesoporous cobalt/nickel LDHs with enhanced electrochemical properties, suitable for energy storage. This method is efficient, straightforward, and environmentally friendly due to the use of ethanol as a solvent (Wang et al. 2024). In contrast, while solvothermal synthesis offers numerous advantages, challenges remain in achieving reproducibility and scalability in real-world applications, necessitating further research to optimize these processes (Yu et al. 2021).

Microwave method

LDH synthesis is an aging technique that involves microwave irradiation. Microwave radiation has the potential to induce aging in a material within a time frame of around 15 to 60 min (Yi et al. 2022). MW modification has the potential to facilitate the generation of homogeneous particles

Fig. 6 Synthesis of hydrotalcite via hydrothermal synthesis process (Chaillot et al. 2021)



(Jubri et al. 2012). The process of reflux maturation leads to the generation of larger particles in comparison to the particles that were originally obtained (Jin et al. 2022b). In order to generate surface regions abundant in Mg–Al LDH, a mixture of magnesium hydroxide ($\text{Mg}(\text{OH})_2$) and aluminum hydroxide ($\text{Al}(\text{OH})_3$) was prepared at a molar ratio of 2:1 (Min et al. 2022; Bai et al. 2023).

The oxide method

The oxide approach employs a trivalent metallic cation configuration in combination with an anion and a divalent metal oxide. A little acidic pH is necessary to permit the controlled breakdown of divalent cation oxide. In order for the procedure to be effective, some prerequisites must be fulfilled. The anion intended for intercalation must possess stability in an acidic environment and have the capability to form a soluble salt with the trivalent cation. Moreover, it is important for the divalent cation oxide to undergo gradual hydrolysis (Jia et al. 2021).

Mechanochemical techniques

Researchers have used mechanochemical processes, such as grinding, milling, or crushing, to produce several LDHs (Zhang et al. 2012; Madhusa et al. 2021). To complete these syntheses, the solid precursors were processed by grinding them in agate or ceramic mortars (Jiang et al. 2020) or crushing them in planetary ball mills. Moreover, several researchers have used this methodology in conjunction with wet synthesis methodologies. Maegawa et al. (2023) used hydrothermal treatment with a ground mixture of reactants to consistently form the LDH phase and enhance its crystallinity (Maegawa et al. 2023). In their study, Ibrahimova et al. (2021) used a mechanochemical methodology to facilitate the conversion of carbonate anions into nitrates by decarbonating the intercalation of anions (Ibrahimova et al. 2021). The specialized wet grinding mill for the continuous batch manufacture of LDHs was built by Barnard and Labuschagné (2020). The synthesis of nanocomposites consisting of LDH materials, namely, Ni–Cr and phosphate buffered saline (PbS), was achieved by the process of grinding and agitating the combination in an aqueous solution (Barnard and Labuschagné 2020).

Numerous instances substantiate the efficacy of this approach in contrast to conventional methodologies. The mechanochemical techniques used to generate yttrium-modified LDH catalysts have shown better reagent activation in the oxidation of cyclohexanone and Claisen-Schmidt condensation compared to catalysts made using coprecipitation (Pavel et al. 2020). Comparable results were obtained when naphtholate AS dye was applied to mechano-Zn LDHs instead of coprecipitated-Zn LDHs (Santosa et al. 2020).

The mechanochemical procedures used to create diclofenac-LDH and its subsequent dispersion in hydroxides demonstrated a higher capacity for anion retention in comparison to conventional coprecipitation, direct anion exchange, and dehydroxylation-rehydration approaches (Szabados et al. 2021). Mechanochemical processes are ecologically sustainable since they only include the segregation and desiccation of the products, leading to little environmental impact.

Table 1 provides a concise overview of various synthesis methods for layered double hydroxides (LDHs), detailing their strengths, limitations, and influencing factors. Each method exhibits unique characteristics that determine its suitability for specific applications and the quality of the resulting LDHs. This summary highlights operational complexity, cost-efficiency, and scalability, making the information readily accessible for further reference or study in industrial and research applications (see Table 1).

CO₂ capture using LDHs

LDHs have significant potential in several fields, including refractive retardation, catalysis, and high-temperature CO₂ collection. Conventional Mg–Al LDHs have attracted considerable investigation, but other compositions have been relatively neglected. Presently, there is a limited understanding of the efficacy of CO₂ capture in LiAl₂ LDH. Research conducted in 2011 to evaluate the CO₂ capture capability of five different kinds of LDHs derived from alkali metal carbonates. The results showed that Li₂CO₃ was the most efficient in capturing CO₂ when integrated into LDHs (Li and Nasiman 2023). Scientists are very interested in manufacturing LiAl₂-LDHs and investigating their potential for adsorbing CO₂. This project entails the synthesis of several LiAl₂ LDHs by the use of coprecipitation and gibbsite intercalation methods, including CO₃, Cl, and NO₃ anions (Sharma et al. 2023).

The produced LDHs were subjected to comprehensive investigation employing techniques including X-ray diffraction (XRD), thermogravimetric analysis (TGA), scanning electron microscopy (SEM), transmission electron microscopy (TEM), Fourier transform infrared spectroscopy (FTIR), BET analysis, and other methods. The research examined the impact of many parameters, including the Li–Al ratio in the preparation solution, calcination temperature, adsorption temperature, and K₂CO₃ doping, on the effectiveness of CO₂ capture across a broad range of temperatures. The stability of CO₂ capture in LiAl₂–CO₃ and K₂CO₃-promoted LiAl₂–CO₃ LDHs was assessed by subjecting them to several cycles of CO₂ adsorption and desorption. This study sought to tackle the worldwide environmental issue of CO₂ emissions by suggesting several CCUS options. An effective approach is improving the conveyance of water

Table 1 Synthesis methods of layered double hydroxides: advantages, disadvantages, and influencing factors

Synthesis method	Advantages	Disadvantages	Key Influential Factors
Co-precipitation method	- Efficient for producing LDH hybrids by integrating biomolecules with exchangeable interlayer anions - Allows control over molecular size and structure through pH maintenance - Facilitates quick and straightforward synthesis with a minimum aging period	- Requires precise control over pH and reactant concentrations to ensure desired outcomes - Temperature and aging duration need careful adjustment to control particle size, which can be a limiting factor in scalability	- pH levels, controlled by titration (e.g., 0.1 N NaOH to maintain pH > 8) - Aging period influences particle size, with shorter periods leading to smaller particles - Temperature manipulation during the process affects the size and distribution of particles - Concentration of reactants and the acidity of the medium impact the efficiency of adsorbents for ion removal
Ion exchange method	- Effective for introducing necessary interlayer anions when co-precipitation or urea methods are inadequate - Greater ease of anion exchange with single-charged host anions compared to multiple-charged ones	- Requires an inert environment to prevent unwanted intercalations, such as carbonate - May involve complex procedures to ensure the effective exchange of anions	- The choice of host anions and their charge characteristics significantly influences the ease and effectiveness of the exchange - Agitation and the concentration of the desired anion in the solution are critical to successful intercalation
Reconstruction (memory effect) method	- Allows regeneration of LDHs while maintaining or enhancing their structural integrity - Capable of integrating guest molecules effectively	- The method requires controlled environments for heating and may involve complex procedures for ensuring the desired outcomes - The process may be less adaptable to large-scale applications due to the stringent control needed over environmental conditions	- Consistent heating rates and specific temperatures are crucial for successful regeneration - The incorporation of guest molecules and the use of templates like carbon microspheres affect the final structure and performance
Urea hydrolysis method	- Produces highly crystalline and pure LDHs - The slow reaction rate of urea decomposition allows for controlled pH elevation and uniform particle formation	- The process can be slow and may generate unwanted byproducts like carbonates, complicating the synthesis - Requires careful handling of urea concentrations and aging processes to maintain product consistency	- Urea concentration and the specific molar ratios of divalent to trivalent metals are crucial - The aging process and synthesis temperature significantly influence the crystallinity and particle size
Sol-gel method	- Produces nanoscale particles with excellent uniformity, purity, and specific surface area - Allows flexibility in modifying the structural characteristics of LDHs by adjusting reactant amounts and aging times - Low-cost method suitable for various applications including medical fields	- Potential decrease in crystallinity, requiring additional techniques like microwave irradiation, hydrothermal treatment, or ultrasonication to enhance quality	- The amount of reactants and the duration of aging directly influence the morphology and structural characteristics - The environmental conditions under which the sol-gel process is conducted (ambient conditions) - The type of metal salts used and their solubility in the chosen solvent
Hydrothermal method	- Enhances crystallinity, purity, and control over particle size of LDHs - Allows for the synthesis under controlled temperature and pressure conditions, beneficial for producing well-defined crystal structures	- Time-consuming and energy-intensive, limiting its scalability and efficiency for large-scale applications	- Temperature and pressure conditions within the reactor are crucial for determining the quality and properties of the synthesized LDHs - The choice of initial materials (e.g., hydroxides, metal oxides) influences the final product characteristics

Table 1 (continued)

Synthesis method	Advantages	Disadvantages	Key Influential Factors
Solvothermal method	- Provides precise control over the size, shape, and composition of the nanostructures, making it ideal for applications in energy storage and other advanced materials - Enables the synthesis of monolayer nanosheets with controlled thickness and stability, critical for high-performance applications - Offers an environmentally friendly alternative due to the use of solvents like ethanol which are less toxic compared to other chemical routes	- Achieving reproducibility and scalability can be challenging, limiting its practical application in large-scale industrial settings - The process can be time-consuming and requires precise control of experimental conditions to ensure consistent results	- The choice of solvent is crucial as it needs to match the surface free energy and solubility parameters of the LDHs to optimize the synthesis - Temperature and pressure conditions within the sealed reactor are vital for the proper formation of LDHs - The type and concentration of alkali sources like urea, which influence the pH and ionic environment during synthesis, play a significant role in the quality of the final product
Microwave method	- Rapid synthesis process due to microwave irradiation, significantly reducing the time required for aging - Facilitates the formation of homogeneous particles, improving the consistency of the LDHs produced	- Potential for uneven heating, which could lead to inconsistent quality across batches - Scalability issues due to the limitations of microwave technology for large-scale synthesis	- Microwave power and exposure time critically influence the aging process and the properties of the final LDH products - The molar ratio of reactants (e.g., Mg to Al) and the types of hydroxides used can affect the synthesis outcome
Oxide method	- Allows for the controlled breakdown of divalent cation oxides, offering precise control over the intercalation of anions - Suitable for environments requiring stability in acidic conditions	- Requires careful pH control and specific conditions to prevent the degradation of the LDH structure - The process complexity might limit its application in less controlled environments	- The pH level is crucial for ensuring the controlled breakdown of divalent cation oxides - The stability of the anion intended for intercalation in acidic environments influences the success of the method
Mechanochemical method	- Enables the production of LDHs through physical processes like grinding, which can be combined with wet synthesis methods to enhance crystallinity - Effective in facilitating the intercalation of different anions through physical manipulation of precursor materials	- Physical force applied during grinding or milling might affect the structural integrity of LDHs - The process might require specific equipment and conditions to achieve desired outcomes	- The intensity and duration of grinding or milling significantly impact the particle size and distribution - The type of mechanical force applied and the properties of the precursor materials determine the efficiency and quality of LDH synthesis

and gas using steam or oxygen-enhanced water–gas shift (SEWGS) processes, which aid in the production of CO₂ and hydrogen. Suggestions have been made for improving materials to enhance the efficiency of CO₂ capture and reaction conversion rates while also reducing emission temperatures and pressures. Researchers have suggested perfluorooctane sulfonate (PFOS) as a suitable substance for capturing CO₂ at moderate temperatures. However, enhancements are required in terms of its ability to adsorb CO₂, its mechanical strength, and its rate of adsorption. Possible strategies for improvement may include researching synthesis methods and internal configurations, developing composite or coated adsorbent materials, or boosting LDH using basic techniques like agitation.

An often-suggested method to enhance LDH adsorption capability of CO₂ is by integrating different quantities of potassium (K), specifically potassium carbonate (K₂CO₃). Integrating LDH with K₂CO₃ could potentially hasten the transformation of MgO into MgCO₃. This occurs because of the presence of carbon compounds, like K-CO₃-Al, which maintain an equilibrium with CO₂. Another suggested approach is to use alkali or alkali metal floor scissors to enhance certain attributes. The research assessed the capacity of treated materials to capture and retain CO₂ at particular temperatures ranging from 50 to 400 °C. The compounds Zn₆Al₂ and Zn₆Ti₂ exhibited the greatest adsorption capabilities, ranging from 0.177 to 0.244 mmol CO₂/g, respectively. The maximal adsorption capacity declined as the temperature rose, ranging from 0.073 to 0.081 mmol CO₂/g. The use of titanium enhanced the CO₂ adsorption capability of the 6Al₂ alloy. This research emphasizes the need to choose cost-efficient and eco-friendly chemicals, advocate for a circular economy approach, and reduce trash disposal to avoid the transfer of hazardous waste to secure storage facilities (Ai et al. 2023).

The M²⁺/M³⁺ ratio in layered double hydroxides (LDHs) is a key factor in their effectiveness in CO₂ capture applications. The ratio of divalent to trivalent cations affects the layer charge density within the LDH structure, affecting the nature and quantity of interlayer anions. These anions are crucial for CO₂ adsorption, as they form stable carbonate complexes. The M²⁺/M³⁺ ratio also affects the LDH surface basicity, structural stability, and thermal resistance. A higher proportion of M²⁺ increases surface basicity, improving interactions with CO₂, enhancing adsorption efficiency. Thermal stability is essential for high-temperature CO₂ capture processes. Optimizing the M²⁺/M³⁺ ratio is crucial for maximizing CO₂ capture capabilities, balancing enhanced surface basicity with structural stability (Kim and Lee 2019; Jerome et al. 2022; Mahinroosta and Allahverdi 2024; Zhang et al. 2025).

As previously said, the use of LDHs for CO₂ capture is receiving significant interest due to their advantageous

properties. The following points provide a comprehensive explanation of various LDH structures and modifications in relation to their uses in CO₂ capture. This comprehensive approach, incorporating LTH formation, calcination, composite formation, alkali impregnation and amine functionalized, offers a promising strategy for maximizing the CO₂ adsorption capabilities of LDH-based materials.

Layered Triple Hydroxides (LTHs)

Different researches were conducted to create new catalysts for technologies that use CO₂ by using copper nanoclusters supported inside the triple hydroxide framework. The objective is to improve the structural configurations and the ability of these catalysts to capture CO₂. The findings suggest that elevated copper concentrations lead to larger crystal volumes while simultaneously decreasing surface areas and pore widths. Copper acts as a dispersant, so the Cu/LTH catalysts have evenly spread out homogeneous nanosheets on the outside of the LTH. The subsequent examination uncovers that the catalyst has the capacity to augment CO₂ adsorption at touch. The Cu₃/LTH catalyst improves the reduction of copper species, specifically Cu⁺. This makes the process of changing CO₂ into methanol more efficient.

Therefore, incorporating improved adsorption capacities into catalysts showcases their practicality and effectiveness in enhancing CO₂ adsorption. Prior to commercial implementation, researchers conducted several early research efforts to improve catalyst qualities, such as altering elemental characteristics and managing microscale structures. Catalyst systems may consist of transition metals and/or precious metals, including zinc (Zn), zirconium (Zr), cerium (Ce), titanium (Ti), platinum (Pt), and others. It is necessary to have these parts in order for the metals to be evenly spread out and for a metal oxide surface to form. This creates better catalytic activity, product selectivity, and performance stability. The recently created Mg-Zn-Al trioxide has attracted considerable interest. The structure of this compound makes it possible for nano-accumulations to form on a variety of exterior surfaces. This helps to enrich areas below the surface and improves the interactions between minerals and the structures that support them. These properties provide benefits in enhancing catalytic performance, perhaps aiding nano-copper clusters in increasing surface area, dispersing metal, extracting CO₂ at high temperatures, and subsequently boosting catalytic performance (Ai et al. 2023).

Calcined LDHs

Heating LDHs at temperatures between 450 and 550 °C may result in the creation of layered double oxides (LDOs). When these LDOs are exposed to aqueous solutions containing anions, water, or a humid atmosphere, their layered

structure can be restored. The moisture-adsorbing capability of LDOs offers a notable benefit in several applications, especially in liquid adsorption. Many scholarly publications explore the process of rehydrating metal oxide compounds to restore LDHs, which is usually referred to as the “memory effect.” Cardinale et al. (2023b) used co-precipitation to create Mg/Al-LDH and transformed it into LDO by calcination at a temperature of 400 °C for 2 h (Cardinale et al. 2023b).

The inclusion of carbonate in the solution enhanced the process of rehydrating low-crystalline LDO (Mayorov and Kopkova 2023). Additionally, Silva Neto et al. (2021) highlighted that drying parameters significantly influenced the physical, structural, and morphological properties of LDH, including crystallite size variations and basal spacing changes (Silva Neto et al. 2021). Mayorov and Kopkova (2023) investigated the process of rehydrating Mg/Al-LDO with a metal molar ratio of 3 (Mg/Al = 3:1) and noticed that the initial pH of the suspension solution had a notable impact on the creation of LDH (Mayorov and Kopkova 2023).

The objective of this work was to examine the rehydration mechanism of LDOs that have undergone high-temperature calcination, especially in the temperature range of 600 to 900 °C (Mayorov and Kopkova 2023). The study used thermal analytical measurements, a mass spectrometer (TG-MS), SEM, infrared spectroscopy (IR), and N₂ sorption isotherms to investigate how different rehydration conditions and durations affect the structure, composition, and specific surface area of the rehydrated LDO. The Mg/Al-LDHs developed were designated as C600-LDO, C700-LDO, C800-LDO, and C900-LDO. TGA, XRD, and Al solid-state MAS NMR investigation using an Agilent VNMRS Direct Drive 400 MHz spectrometer, and diffuse reflectance infrared Fourier transform (DRIFT) spectra were used to identify the composition of the materials. It is important to consider the impact of treatment parameters, such as temperature and hydration conditions, on the fluctuation of a specific surface area. The surface area can be measured using a multipoint BET analysis.

The synthesis of MgAl-LDH by co-precipitation was also reported in the literature. The produced MgAl-LDH was analyzed using SEM with an energy-dispersive X-ray (EDX) in order to probe its morphology and composition. The analysis confirmed that the ratio of Mg²⁺ to Al³⁺ was as predicted, with a value of 3. A study using XRD patterns revealed that the LDO changed into LDHs (re-LDHs) very quickly (partially). This happened for 6 h. High-temperature calcination triggered this transformation in all LDO samples. This discovery diverged from the rehydration procedures described in the literature, which typically include using a Mg/Al-LDH with a Mg/Al ratio of 2 (Mayorov and Kopkova 2022).

The LDO calcination temperature and rehydration parameters, such as water phase and duration, had an impact on

the rehydration process and the degree of moisture recapture. When rehydrating samples of C600-LDO, C700-LDO, C800-LDO, and C900-LDO in the presence of water vapor and liquid water, the XRD examination revealed incremental changes in reflection intensity and clarity. Nevertheless, the presence of liquid water resulted in much more distinct reflections within a time frame of 6 h. Rehydration was more rapid in liquid water than in vapor, resulting in the formation of Re-12 LDH crystals at a higher rate in the former. A crystal size increase was seen only in C800-LDO and C900-LDO samples that were rehydrated in liquid water. When calcined at temperatures of 700 and 800 °C, followed by rehydration in liquid water rather than vapor, samples subjected to rehydration for 24 and 48 h showed greater intensity and more distinct features in their XRD reflections. Rehydration in liquid water at a temperature of 900 °C resulted in a decrease in XRD reflections compared to rehydration in vapor. The research showed that early LDOs had the capacity to completely rehydrate and transform into LDHs, with the pace of rehydration being influenced by temperature and slowing down at higher temperatures. The rehydration procedure resulted in variations in both the size and shape of the crystals.

The lattice parameters of the LDOs rose from 3.05 to 3.11 Å after being calcined at 900 °C. Small changes in the Mg/Al molar ratio likely caused this change. Moreover, the study examined the process of alumina rehydration and the resulting materials, with a particular focus on the degree of crystallinity and the progression of their formation. The process of rehydration was more rapid and comprehensive when conducted in liquid water as opposed to vapor. Additionally, the size of the crystals decreased as the temperature for calcination increased. Nevertheless, when the rehydration periods were reduced, the crystal sizes remained about the same in all samples that were exposed to different calcination temperatures. This indicates that the presence of liquid water speeds up the process of crystal formation. The solid-state MAS NMR spectra of freshly synthesized LDH, LDOs at different temperatures, and rehydrated C900-LDOs showed that the LDH phase was present, with a peak that could be easily seen at 10 ppm. The rehydrated samples had an extra tetrahedral peak at 69.2 ppm, which is probably due to the presence of the spinel (MgAl₂O₄) phase. To summarize, our study highlights the significance of rehydration in attaining greater crystal diameters of rehydrated LDHs and facilitating crystal growth (Wei et al. 2022a). Figure 7 illustrates the structural arrangement of Mg/Al-LDH and Mg/Al-LDO. Figure 8 illustrates the structural alterations in MgAl-LDH after calcination, showing the stability and alignment of Mg²⁺ and Al³⁺ ions. It also depicts the thermal decomposition of LDH during calcination to yield mixed metal oxides, detailing the release and transformation of carbonate ions.

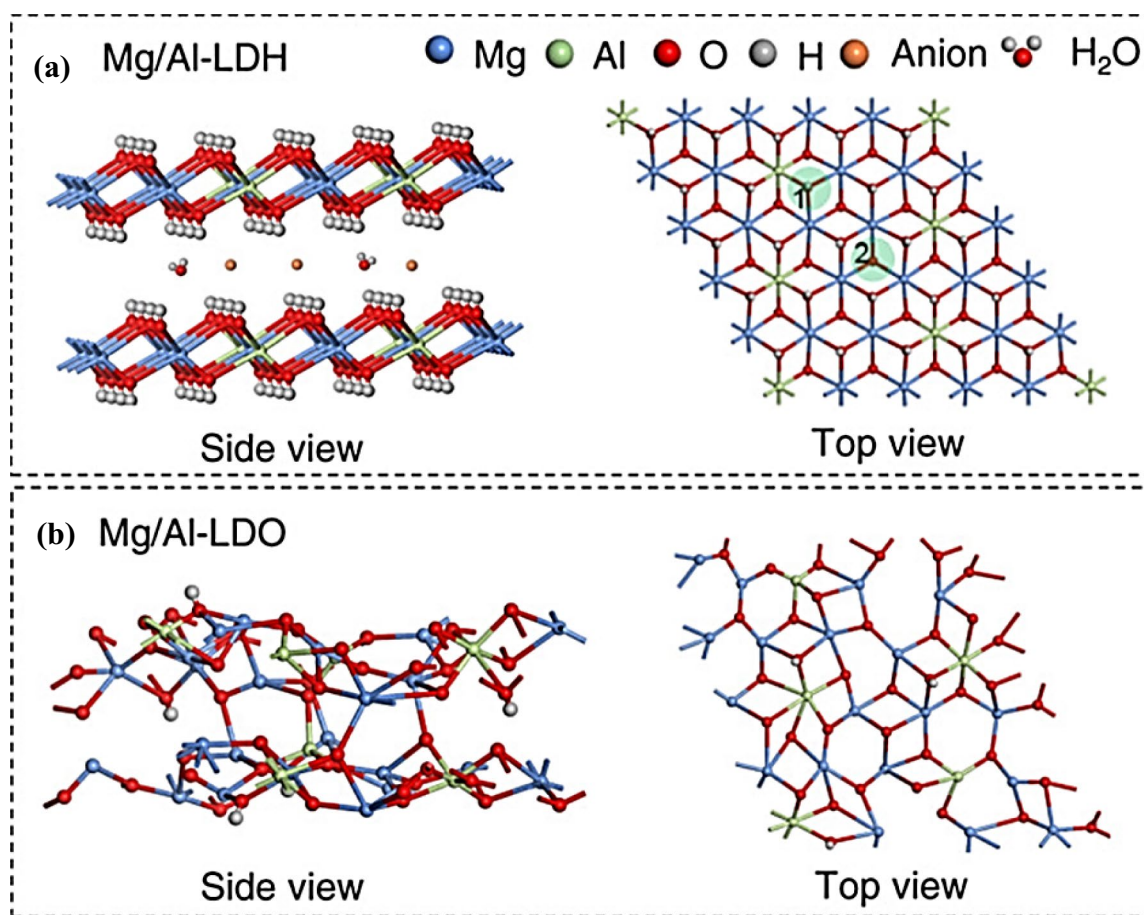


Fig. 7 Structural configuration of **a** Mg/Al-LDH and **b** Mg/Al-LDO (Jin et al. 2022a)

CO₂ capture using composites containing LDHs

A material containing LDHs is often characterized by the integration or mixing of LDHs with another component, resulting in the formation of a composite structure. The use of LDH composites presents improved mechanical properties, increased thermal stability, and the possibility of providing specific functionalities such as regulated ion or gas distribution (Garcia-Gallastegui et al. 2012a; Wang et al. 2018; Fernández et al. 2020). Research is now being conducted on composites that include LDHs, specifically LDH/carbon nanotube (LDH/CNT) and LDH/graphene oxide (LDH/GO) hybrids, to enhance the efficacy of CO₂ capture. LDHs possess a structural arrangement with an anionic layer, which facilitates the incorporation of various functional molecules. The incorporation of LDHs into composites enhances the capacity for CO₂ adsorption and selectivity (Wang et al. 2016). In general, the process of creating nanocomposites may be classified into two primary methods. One is a one-pot co-precipitation technique. Another method of synthesis involves the self-assembly of positively charged delaminated LDH single

sheets and negatively charged monolayers, which is triggered by electrostatic forces (Wang et al. 2016).

The synthesis of an LDH/GO hybrid material for CO₂ capture was conducted by Garcia-Gallastegui et al. (2012a) using the co-precipitation technique. The researchers observed an increase in the CO₂ sorption capacity and recyclability when LDHs were supported on GO, which may be attributed to the better dispersion of particles. Their study pinpointed the most effective GO concentration for maximum efficiency to be between 5 and 10 wt.%, a finding that is visually represented in Fig. 9. This study examines the interaction between LDH and GO, focusing on their structural relationship. Additionally, it analyzes the performance of LDH and GO in adsorbing CO₂ under certain conditions: a temperature of 573 K and a CO₂ partial pressure of 0.2 bar (Garcia-Gallastegui et al. 2012a).

The nanocomposite adsorbent was generated and evaluated by Wang et al. (2015) by electrostatic self-assembly between a single sheet of positively charged Mg–Al LDH and a monolayer of negatively charged GO. Prior to each CO₂ adsorption test, the Mg–Al-NO₃ LDH and Mg–Al-NO₃ LDH-NS/GO nanocomposites underwent calcination at 400

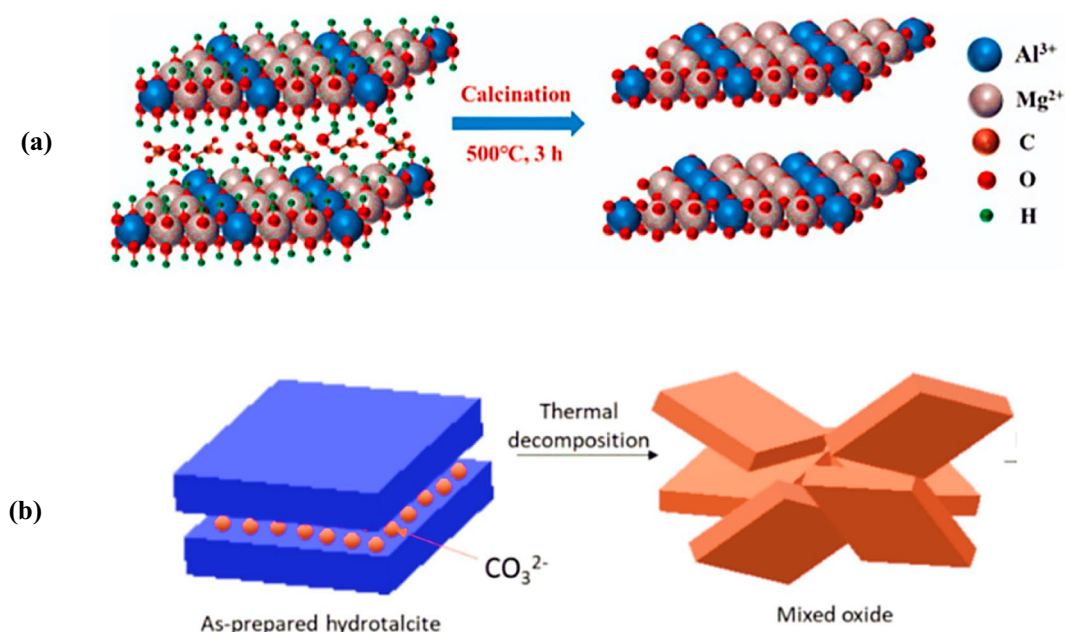
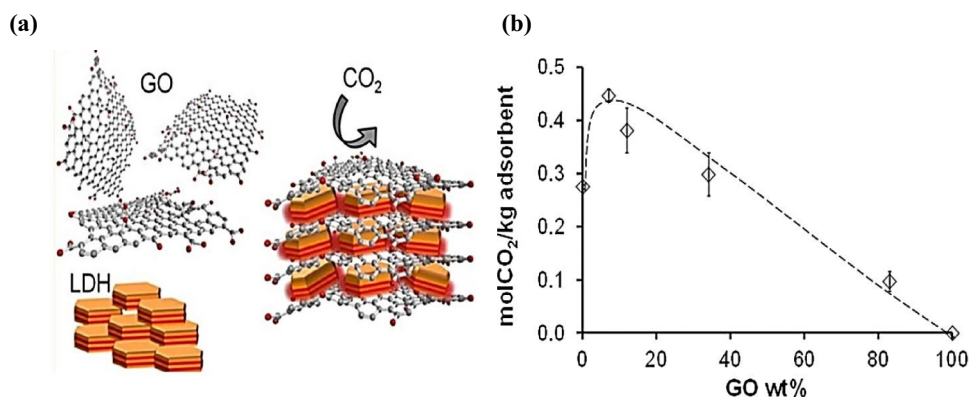


Fig. 8 **a** Illustrates the alterations in the structure of MgAl-LDH after undergoing calcination, as reported by Long et al. (2021) and **b** the thermal decomposition of LDH during calcination to yield mixed metal oxides (Lauermannová et al. 2020)

Fig. 9 **a** Structural combination and **b** performance of LDH/GO (Garcia-Gallastegui et al. 2012a)



°C for 5 h. A thermogravimetric study was performed in order to evaluate the sorption of CO₂ on nanocomposites consisting of Mg–Al–NO₃ LDH and Mg–Al–NO₃ LDH-NS/GO. The research showed that the addition of GO enhanced the CO₂ adsorption capacity and multi-cycle stability of LDH via improving particle dispersion and stabilization. The use of 6.54 wt.% GO as a support material resulted in a more than twofold increase in the CO₂ capture capacity of LDH (Wang et al. 2015).

Wang et al. (2016) investigated the impact of the manufacturing method and chemical composition on the CO₂ capture capacity of LDH/oxidized carbon nanotube (LDH/OCNT) nanocomposites. The use of OCNT in nanocomposites of LDH demonstrates remarkable stability during CO₂ adsorption and desorption cycles. Additionally, these

composites display improved pelletization properties compared to pure LDH while also providing superior mechanical and thermal stability. The results of the investigation suggest that the LDH-NO₃-NS/OCNT nanocomposite has enhanced adsorption capacity in comparison to the OCNT-free composite. The LDH-NO₃-NS/OCNT nanocomposite, which consisted of 9.1 wt.% OCNT, exhibited the greatest adsorption capacity compared to pure Mg–Al–NO₃ LDH, surpassing it by a factor of more than two. As a result, they have significant potential for CO₂ capture and mitigation (Wang et al. 2016).

A unique set of DAC adsorbents was developed by the application of branching PEI onto MMOs generated from Mg_xAl-CO₃ LDH. The composites are produced using a cost-effective and simple production method. PEI/MMO

composites have exceptional thermal and hydrothermal resilience, making them very suitable for the efficient collection of CO₂ on a broad scale, even at exceedingly low concentrations. Also, PEI/MMO composites exhibit crucial characteristics, including the ability to retain a porous structure even when loaded with large amounts of PEI. The uniform distribution of impregnated PEI across the surface of the support material improves the accessibility of CO₂ molecules to active adsorption sites and expedites the rate of CO₂ adsorption. This leads to a combined CO₂ adsorption of 2.27 mmol/g and a fast adsorption rate of 1.1 mmol/g/h over a 90-min timeframe at 25 °C with a CO₂ pressure of 0.4 mbar. The appropriateness of the hybrid adsorbents for real-world CO₂ capture applications was shown by a minimum of 20 cycles of testing under both dry and humid environments, hence assessing their durability. The presence of moisture enhances the capacity of materials to adsorb CO₂ and prolongs their lifespan, hence augmenting their efficacy in adsorbing CO₂ in practical situations. These findings indicate that the PEI/MMO composites have promising characteristics as efficient and cost-effective adsorbents for the capture of CO₂ at very low concentrations (Zhu et al. 2020b).

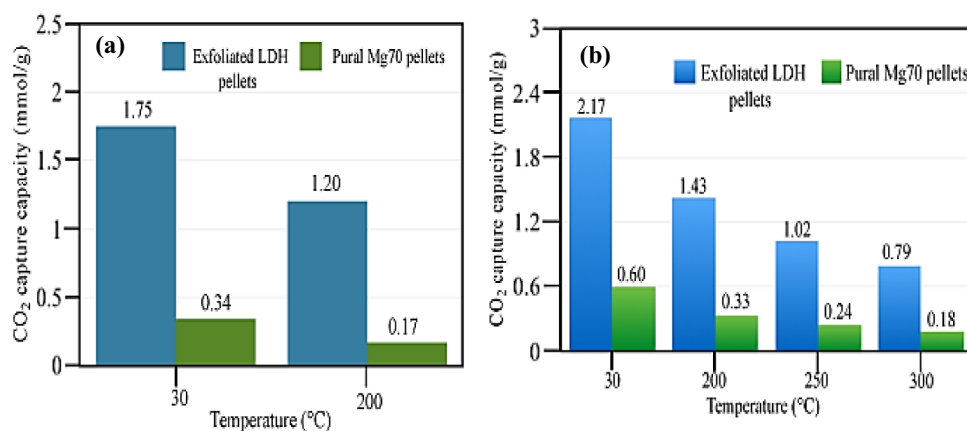
An extensive study has been conducted on the use of LDHs in conjunction with MMOs as sorbents to CO₂ in various applications that require elevated temperatures ranging from 200 to 400 °C (Valizadeh et al. 2018; Cousin-Saint-Remi et al. 2019; Veerabhadrapa et al. 2021). However, to be acceptable for large-scale CO₂ capture in industries, it is necessary to improve the CO₂ adsorption efficiency of LDHs and related MMOs in terms of both capture capacity and cycle stability. The research used a modified amide hydrolysis method to improve the ability of LDHs obtained from MMOs to adsorb CO₂. Aqueous exfoliation and freeze-drying procedures were used to create novel MMOs based on LDHs. The methodology involves the exfoliation of LDH nanosheets followed by their pelletization in order to overcome the challenges associated with using powdered samples in industrial applications on a large scale. The exfoliated

freeze-dried pellets exhibited improved levels of CO₂ capture at both room temperature 30 °C (1.75 mmol/g under 14% CO₂, 2.17 mmol/g, under 86% CO₂) as well as medium temperatures 200 °C (1.2 mmol/g under 14% CO₂, 1.43 mmol/g under 86% CO₂) in comparison to the first powder samples of MMOs as depicted in Fig. 10 (Veerabhadrapa et al. 2021).

Wu et al. (2022a) investigated the use of (xLDH/mesoporous alumina, *x* is the loading of LDH) composites for CO₂ capture (Wu et al. 2022a). Alumina has received considerable focus owing to its high surface area, voluminous pore capacity, rich mesoporous configuration, superior crystallinity, and excellent thermal and mechanical durability (Chen et al. 2014; Gunathilake et al. 2016; Wu et al. 2022a). The xLDH/MA composites were found to possess a large BET surface area (278–378 m²/g), narrow pore-size distribution (4.37–6.23 nm), and a highly crystalline framework and exhibited good regeneration performance, with a regeneration efficiency of more than 89% after 11 recycles, showing potential for application in CO₂ capture. Among the samples, 30LDH/MA composite material showed fast adsorption kinetics and good regeneration efficiency and exhibited the highest CO₂ adsorption capacity 1.56 mmol/g at 50 °C, which was 1.55 times larger than that of unmodified MA making it a promising candidate for CO₂ capture applications (Wu et al. 2022a).

Wang et al. (2023a) investigated the use of NiFe-LDH/halloysite composites for the purpose of CO₂ capture. The composite materials were fabricated using hierarchical structures in order to provide a significant surface area for the adsorption of CO₂. The use of halloysite as a substrate for NiFe-LDHs in CO₂ adsorption processes offers notable advantages. The object has a unique cylindrical form with a substantial surface area, enabling a notable active surface for the adsorption of CO₂ shown in Fig. 11a. This characteristic makes them suitable candidates for the development of sustainable carbon capture technology. This provides efficient and environmentally friendly methods for addressing

Fig. 10 The CO₂ capture capabilities of exfoliated and commercial LDH-based MMOs at atmospheric pressure and varying temperatures. The measurements were conducted under two conditions: **a** with 14% CO₂ and **b** with 86% (Veerabhadrapa et al. 2021)



challenges related to carbon capture. The sample of NiFe-LDH/Hal600 exhibited a remarkable capacity for adsorbing CO₂, which may be attributed to the substantial interaction between the surface and CO₂ molecules. The composites exhibited the ability to undergo recycling and maintained their stability across several cycles as shown in Fig. 11b (Wang et al. 2023a). Table 2 offers a comparative analysis of the CO₂ adsorption capacity exhibited by different adsorbents.

CO₂ capture using alkali-impregnated LDHs

LDHs are materials that have been altered by the addition of alkali metal ions. Introducing alkali metals may change the structural, chemical, and adsorption characteristics of LDHs. This alteration may improve the material capacity to adsorb CO₂ by facilitating the interaction between alkali metal ions and CO₂ molecules, creating dense and evenly distributed surface sites (Yang et al. 2021; Tan et al. 2024).

The role of potassium in the adsorption process is being examined (Santamaría et al. 2023). LDH was treated with K₂CO₃ to enhance its adsorption capability. The method involving elevated supersaturation levels with a 2:1 ratio of Mg to Al demonstrated the most favorable adsorption and desorption characteristics. Optimal impregnation with K₂CO₃, at a concentration of 20 wt.%, resulted in a maximum adsorption capacity of 0.77 mmol CO₂/g under conditions of 450 °C and 800 mmHg (Yang and Kim 2006). Impregnating potassium carbonate to calcined LDH samples enhances their adsorption capacity (Santamaría et al. 2023). During the CO₂ adsorption process, a significant quantity of K₂CO₃ was involved in the reaction, leading to the formation of both monodentate and bidentate carbonates on K₂CO₃/LDH, as illustrated in Fig. 12. This indicates a mixture of various carbides and suggests that potassium plays a role in facilitating a chemical reaction between the sample and CO₂ (Li et al. 2024b).

However, using different solvents may significantly alter their performance. Zhu et al. (2020a) explored the process of infusing exfoliated Mg₃Al–CO₃ LDH nanosheets with a 20 wt.% concentration of K₂CO₃ using a vacuum

procedure and six different solvents. The results of the investigation revealed that the choice of solvent utilized during the impregnation process had a significant impact on the sample adsorption capacity. The investigation encompassed the evaluation of six impregnation solvents shown in Fig. 13, among which ethanediol emerged with the highest CO₂ uptake capacity, measuring 1.46 mmol/g at 400 °C for 20 wt.% K₂CO₃ promoted CC1. Analysis of material properties suggests that the superior performance of K₂CO₃/CC1 (ethanediol) can be ascribed to the uniform distribution of K⁺ ions on the surface and the elimination of residual solvent during calcination, thereby facilitating CO₂ diffusion (Zhu et al. 2020a).

Walspurger et al. (2010) propose that enhancing LDH with K₂CO₃ could promote the carbonation of periclase (MgO). At 350 °C, a hydrotalcite-based material promoted with 11 wt.% K₂CO₃ and Mg/Al ratio of 2.9 can hold a maximum of 15.1 mmol/g of CO₂ when exposed to adequate levels of CO₂ and steam (Walspurger et al. 2010). The research conducted by Li et al. (2013b) introduced a novel precursor, stearate-pillared LDH enhanced with K₂CO₃, for CO₂ adsorption purposes with general schemes shown in Fig. 14. Experimental outcomes from a TGA demonstrated an impressive CO₂ capacity of 1.93 mmol/g at 300 °C, which is 1.7 times greater than that of a conventional K₂CO₃-promoted hydrotalcite adsorbent (1.11 mmol/g), along with rapid adsorption kinetics. Although there are some small impurities and acid damage, this precursor, which involves the intercalation of stearate anion and the enhancement of K surface, has significant promise for capturing CO₂ before combustion and for industrial manufacturing (Li et al. 2013a).

Zhu et al. (2019a) achieved a successful synthesis of AMO-LDHs with varying Mg/Al ratios by utilizing a novel technique called AMOST (aqueous miscible organic solvent treatment). This particular method involved the treatment of LDHs with AMO solvent, which was discovered to offer additional alkali ion promoters for surface modification, ultimately resulting in enhanced dispersion. After impregnation of 20 wt.% K₂CO₃, K-LDOs exhibited positive correlations between CO₂ working capacities and

Fig. 11 **a** Schematic diagram of the NiFe-LDH/Hal composites, **b** CO₂ capture efficiency across 25 cycles from adsorption–desorption isotherms at 298 K and 1 bar (Wang et al. 2023a)

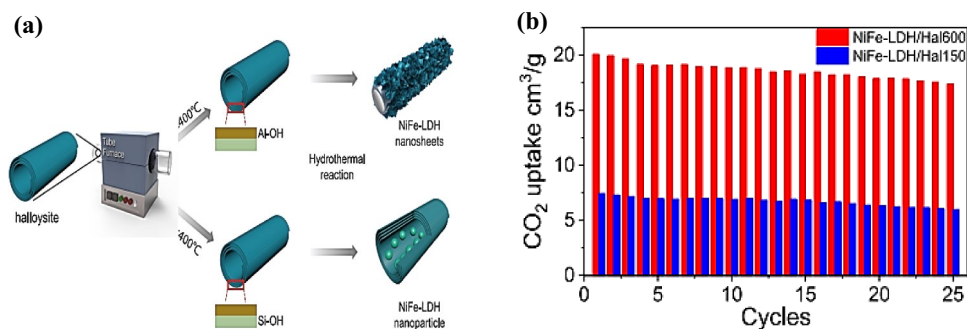


Table 2 Presents a comparative analysis of the CO₂ adsorption capacity shown by several adsorbents

LDH	Temperature °C	Pressure atm	Capacity mmol/g	References
Mg–Al–NO ₃ LDH	200	1	0.24	Wang et al. (2015)
Mg–Al–NO ₃ LDH-NS/GO (1:10)	200	1	0.16	Wang et al. (2015)
Mg–Al–NO ₃ LDH-NS/GO (1:5)	200	1	0.30	Wang et al. (2015)
Mg–Al–NO ₃ LDH-NS/GO (1:1)	200	1	0.38	Wang et al. (2015)
Mg–Al–NO ₃ LDH-NS/GO (1:0.7)	200	1	0.47	Wang et al. (2015)
Mg–Al–NO ₃ LDH-NS/GO (1:0.5)	200	1	0.42	Wang et al. (2015)
Mg–Al–NO ₃ LDH-NS/GO (1:0.3)	200	1	0.32	Wang et al. (2015)
Mg–Al LDH–NO ₃	200	1	0.24	Wang et al. (2016)
Mg–Al LDH–CO ₃	200	1	0.46	Wang et al. (2016)
Mg–Al LDH–CO ₃ /GO	300	200 mbar	0.45	Ferrer (2016)
Mg–Al LDH–NS–NO ₃ /OCNT	200	1	0.43	Wang et al. (2016)
Mg–Al LDH–NO ₃ /OCNT	200	1	0.43	Wang et al. (2016)
Mg–Al LDH–CO ₃ /OCNT	200	1	0.45	Wang et al. (2016)
Mg–Al LDH–CO ₃ /OCNT	300	-	0.42	Garcia-Gallastegui et al. (2012b)
67%PEI-MMO Mg–Al LDH	25	0.4 mbar CO ₂	2.27	Zhu et al. (2020b)
50%PEI-MMO Mg–Al LDH	25	0.4 mbar CO ₂	1.66	Zhu et al. (2020b)
Mg–Al LDH–MMO (14% CO ₂)	30	1	1.75	Veerabhadrapa et al. (2021)
Mg–Al LDH–MMO (86% CO ₂)	30	1	2.17	Veerabhadrapa et al. (2021)
Mg–Al LDH–MMO (14% CO ₂)	200	1	1.2	Veerabhadrapa et al. (2021)
Mg–Al LDH–MMO (86% CO ₂)	200	1	1.43	Veerabhadrapa et al. (2021)
Mg–Al LDH	50	1 bar	0.82	Wu et al. (2022a)
30 Mg–Al LDH/MA	50	1 bar	1.56	Wu et al. (2022a)
10 Mg–Al LDH/MA	50	1 bar	1.01	Wu et al. (2022a)
20 Mg–Al LDH/MA	50	1 bar	1.32	Wu et al. (2022a)
40 Mg–Al LDH/MA	50	1 bar	1.38	Wu et al. (2022a)
AMO- Mg–Al LDH	25	1 bar	1.05	Zhu et al. (2021a)
NiFe-LDH	40	1 bar	3.9 ± 0.2 ^a	Wang et al. (2023a)
NiFe-LDH/Hal150	40	1 bar	6.1 ± 0.4 ^a	Wang et al. (2023a)
NiFe-LDH/Hal 300	40	1 bar	9.0 ± 0.7 ^a	Wang et al. (2023a)
NiFe-LDH/Hal400	40	1 bar	13.6 ± 0.4 ^a	Wang et al. (2023a)
NiFe-LDH/Hal600	40	1 bar	26.9 ± 1.1 ^a	Wang et al. (2023a)
NiFe-LDH/Hal800	40	1 bar	20.2 ± 1.5 ^a	Wang et al. (2023a)

^aReported CO₂ adsorption capacity is in the unit of mg/g

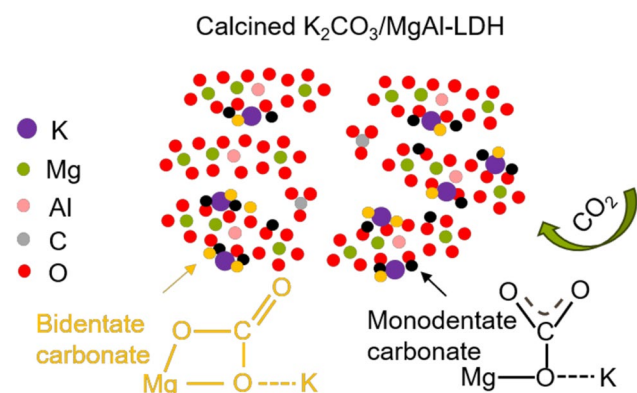


Fig. 12 Illustrative diagram of CO₂ adsorption processes in K₂CO₃/Mg–Al-LDH (Li et al. 2024b)

the total quantity of exposed Mg–O sites. After subjecting the K-Mg₃Al-CO₃ to AMO solvent treatment and performing 10 cycles of testing, the CO₂ working capacity stabilized at 1.069 mmol/g at 400 °C, surpassing the performance of both K-Mg₃Al-CO₃-Sasol(c) and K-Mg₃Al-CO₃-w(c) (Zhu et al. 2019a).

Extensive research has been conducted to determine the ideal quantity of K₂CO₃ for best outcomes, since an excessive amount might block pores and active sites, resulting in decreased adsorption (Santamaría et al. 2023). Bhatta et al. (2015) and Wang et al. (2015) investigated the CO₂ sorption ability of LDHs by doping them with K₂CO₃, which enhances the alkalinity of sorbents, which is beneficial for adsorbing acidic CO₂. Samples with different concentrations of K₂CO₃ were created, and the mixture was dehydrated

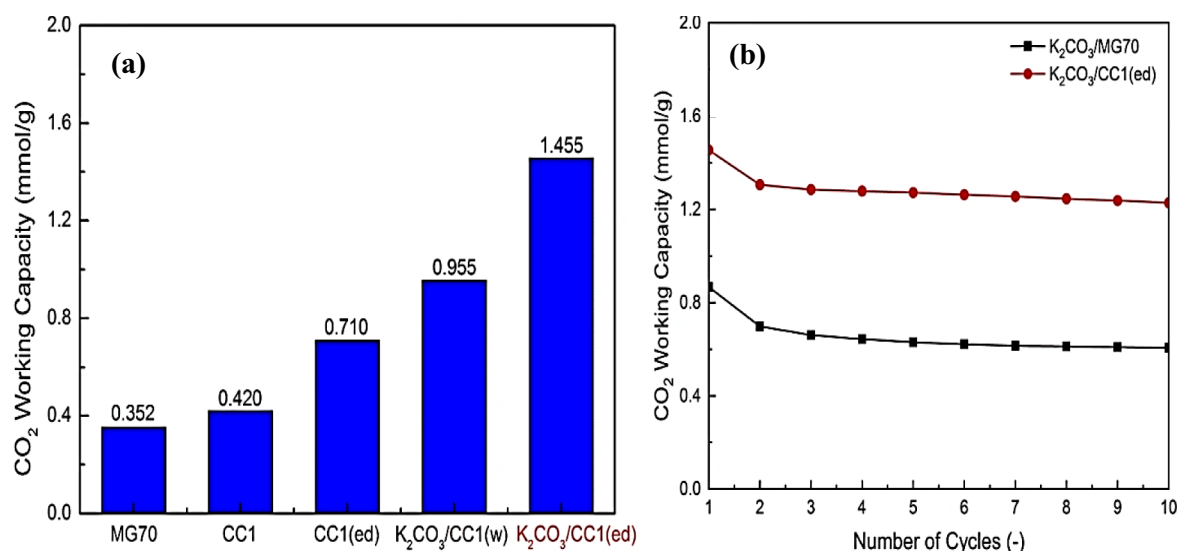


Fig. 13 CO₂ working capacities of samples: **a** initial cycle and **b** after 10 cycles (Zhu et al. 2020a)

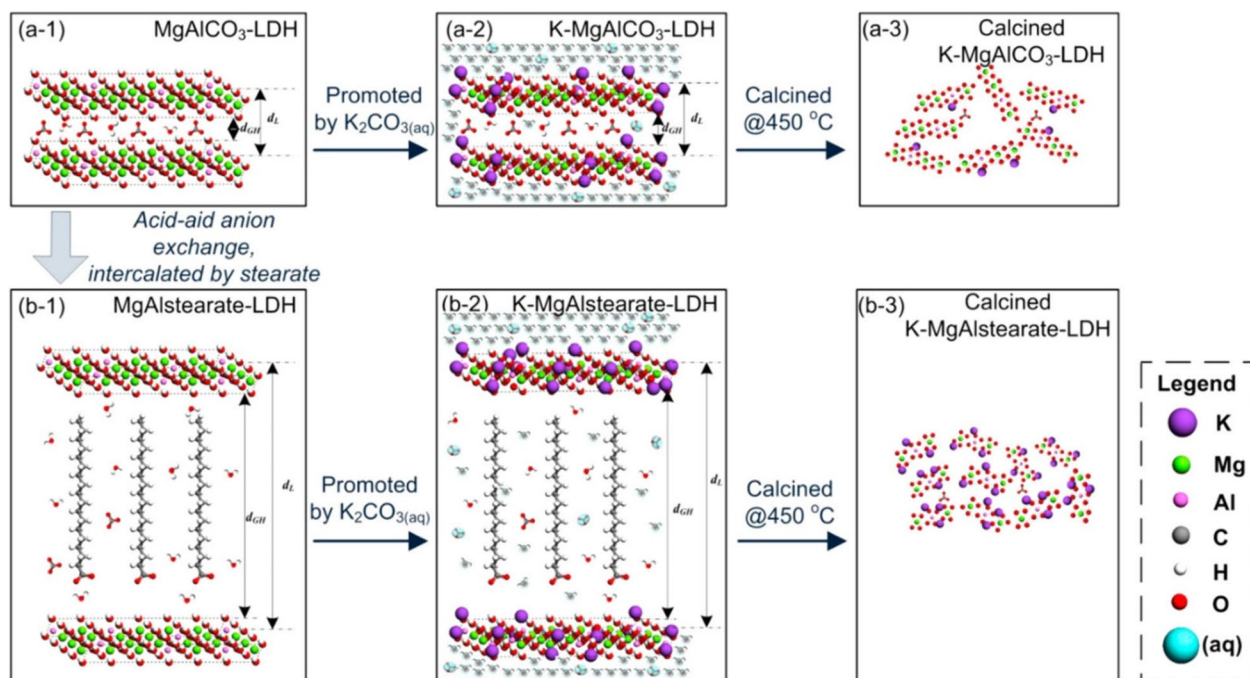


Fig. 14 Transformation of LDH structures through K⁺ ion exchange and calcination at 450 °C for **a** MgAlCO₃-LDH, and **b** MgAlstearate-LDH (Li et al. 2013a)

and re-calcined before conducting CO₂ capture experiments. The results showed that the CO₂ adsorption capability increased with K₂CO₃ concentration from 5 to 15 wt.%, with a maximum CO₂ uptake capacity of about 0.6 mmol/g (Wang et al. 2015; Bhatta et al. 2015).

Limited studies have focused on the application of various potassium compounds to boost the adsorption efficiency of LDH (Santamaría et al. 2023). The investigation conducted

by Sun et al. (2020) delved into the effects of Na/KNO₃ and K₂CO₃ on the adsorptive properties of commercial hydrotalcite MG60, characterized by a Mg/Al molar ratio of 1.91. The presence of Na/KNO₃ molten salt within the micropores of hydrotalcite, owing to its relatively low melting point of 493 K, facilitates the diffusion of CO₂, thereby improving the kinetics of CO₂ adsorption and desorption at temperatures exceeding 493 K. Enhancements in CO₂ adsorption

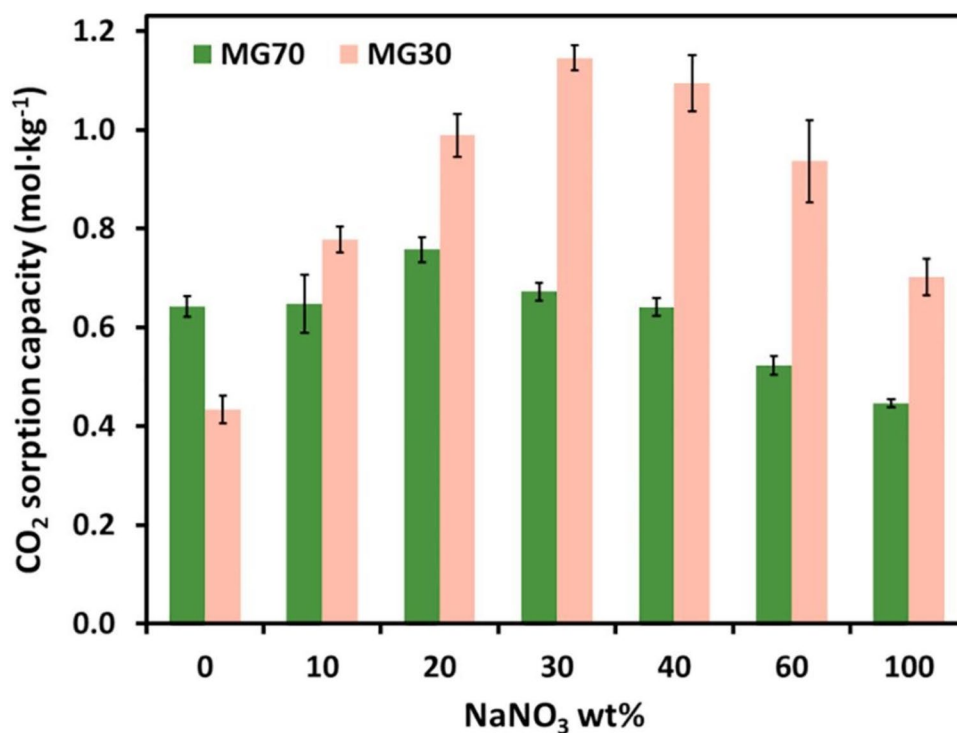
capacity were observed for both types of modified hydrotalcites, with Na/KNO₃ contributing to an increase in alkalinity and K₂CO₃ leading to the formation of K₂Mg (CO₃)₂. This resulted in peak adsorption capacities of 1.07 mmol/g and 1.19 mmol/g, respectively. While both impregnated salts improved CO₂ adsorption, Na/KNO₃-impregnated hydrotalcites demonstrated better performance at 473–573 K, whereas K₂CO₃-impregnated ones exhibited superior performance at 573–673 K, addressing different temperature ranges in industrial CO₂ capture processes making it more suited for SEWGS operational requirements (Sun et al. 2020; Santamaría et al. 2023).

Sodium carbonate can enhance CO₂ adsorption in calcined LDH, a cost-effective alternative to potassium carbonate (Santamaría et al. 2023). Min et al. (2014) discovered that NaNO₃ and MG70 had the highest performance when using a Na₂CO₃/LDH ratio of 0.35:1. LDH specific surface area reduced after impregnation with Na₂CO₃, although increased alkalinity improved their adsorption capacity. With rising temperatures, the Na₂CO₃ crystal structure transitioned from monoclinic to hexagonal, converting CO₂ into carbonate (Min et al. 2014). Studies indicate that the presence of NaNO₃ on mixed oxide surfaces, particularly those with a high Mg/Al ratio, facilitates the creation of MgO sites and carbonates, which are crucial for the development of MgCO₃ crystals. Nevertheless, a high concentration of NaNO₃ might obstruct pores and diminish the surface area available for chemical reactions (Kim et al. 2016a). Hanif et al. (2019) found that impregnation with

NaNO₃ improves adsorption capacity by 28% and lowers energy penalty (Hanif et al. 2019). Doping with alkali metals such as NaNO₃ is recognized as an effective strategy for increasing the CO₂ adsorption capabilities of hydrotalcite, introducing an innovative method to strengthen its basic properties. In a study conducted by Kim and Lee (2019) NaNO₃ was utilized as a precursor to increase the basicity of commercial hydrotalcite by including alkali metals. The research discovered a notable increase in the ability to adsorb CO₂, specifically in hydrotalcite compositions including 30 wt.% NaNO₃ inside a framework consisting of mixed MgO and Al₂O₃. It demonstrated the greatest CO₂ adsorption capacity, reaching 1.15 mol/kg at 200 °C and 1 atmosphere shown in Fig. 15. Additionally, the degree of enhancement in CO₂ sorption uptake was observed to vary based on the Mg/Al molar ratio, with higher levels of Al₂O₃ contributing to increased basicity and sorption performance (Kim and Lee 2019). In a previous study, it was observed that hydrotalcites impregnated with 35 wt.% K₂CO₃ (MG30, MG50, and MG70) had CO₂ sorption uptakes of 0.65–0.90 mol/kg at 200 °C, lower than those treated with NaNO₃ (Lee et al. 2010).

Furthermore, other metals, such as cesium and strontium, underwent testing as modified material (Tan et al. 2024). Oliveira et al. (2008) investigated the adsorption ability of three commercially available LDH samples, namely, MG30, MG50, and MG70, which were impregnated with potassium and cesium at a concentration of 20% by weight. The findings indicated that MG30-K exhibited the highest sorption

Fig. 15 CO₂ adsorption capacity of untreated hydrotalcite and hydrotalcite impregnated with sodium nitrate (NaNO₃) (Kim and Lee 2019)



capacity at 0.76 mol/kg, while MG70-Cs demonstrated the highest capacity among the cesium-impregnated samples at 0.41 mol/kg, under the specific adsorption conditions of 400 °C, 0.4 bar CO₂ partial pressure, and steam. Furthermore, MG30-K displayed a slight loss in capacity (7%) after 75 cycles during cyclic stability testing, suggesting its potential suitability for practical applications (Oliveira et al. 2008). Miguel et al. (2014) discovered that the efficiency of Mg/Al + Ga LDH was enhanced when modified by impregnation with potassium, cesium, or strontium. Potassium demonstrated the most notable outcomes, particularly for HTC-10Ga-20K, which exhibited an exceptional capacity of 2.01 mmol/cm³ at 1.08 bar, followed by cesium and strontium (Miguel et al. 2014).

Faria et al. (2022) analyzed the effects of varying concentrations of Na or Cs in Mg–Al LDHs through wet-impregnation, contrasting these with MMO, K-20 wt.%, enriched MMO, and the commercially available MG63. Their analysis included both the maximal sorption capacity at equilibrium and the capacity after multiple cycles. It was found that achieving the best performance needs a little impregnation % to prevent pore blockage. Out of all the materials that were examined, the K-MMO had the greatest ability to adsorb and retain substances at a rate of 0.67 mmol/g. However, the slower pace at which it adsorbs might hinder the effectiveness of the separation process. The MMO sample with a 10% weight of Cs had the best performance during cyclic operation at a CO₂ partial pressure of 0.15 bar and a temperature of 573 K (Faria et al. 2022).

The provided data in Table 3, offers a comprehensive look at the CO₂ adsorption capacities before and after treatment across various conditions and materials. Analyzing the table reveals several key patterns and trends that are crucial for understanding the efficiency of different alkali-impregnated materials in CO₂ capture. Firstly, the increase in CO₂ adsorption capacity after treatment is evident across all conditions, highlighting the effectiveness of alkali impregnation in enhancing CO₂ capture. Notably, the use of materials like Mg₂Al-CO₃ and its variants, when impregnated with different percentages of alkali (ranging from 10 to 35.35%), shows a significant improvement in adsorption capacity after treatment, illustrating the positive impact of alkali concentration on CO₂ capture efficiency. Temperature and pressure conditions also play a crucial role in the adsorption process. For instance, materials treated at higher temperatures (up to 400 °C) and under varying pressure conditions (from 0.15 to 24 bar) demonstrate varied adsorption capacities, indicating that both temperature and pressure are critical factors in optimizing the CO₂ adsorption process. Moreover, the introduction of different metals and modifications, such as Ga substitution in Mg₂Al-CO₃ or the addition of NO₃/GO nanosheets and stearate, further influences the CO₂ adsorption capacity. This suggests that the structural

and compositional modifications of the adsorbent materials can significantly affect their CO₂ capture capabilities. In summary, the reported findings underscore the importance of alkali impregnation, temperature, pressure, and material composition in enhancing CO₂ adsorption capacities. These findings provide valuable insights into the optimization of materials for CO₂ capture, pointing towards the potential for tailored approaches based on specific operational conditions and material characteristics.

CO₂ capture using amine-functionalized LDHs

The effective and rapid adsorption of CO₂ from the surrounding air is achieved by the use of amine-functionalized, water-miscible organic Mg_xAl-CO₃ LDH (AMO-LDH) nanosheets. The surface of exfoliated AMO-LDH-derived nanosheets was modified by attaching aminosilane monolayers, which were then used in the first instance of DAC. The interaction between LDHs and aminosilanes is essential for the capture of CO₂ as it enables the formation of nanosheets grafted with amine monolayers, resulting in the generation of adsorbents with high CO₂ capacities, quick kinetics, and great sorption–desorption stability as illustrated in Fig. 16. This process improves the availability of amine-active sites for the adsorption of CO₂. In simulated air, the adsorbent exhibited a CO₂ capacity of 1.05 mmol/g and after 30 min, the CO₂ adsorption reached 70% of this maximum capacity. Remarkably, a negligible decline in performance was seen after 50 cycles, indicating the long-lasting nature of the modified LDHs for the purpose of CO₂ capture. The findings underscore the efficacy and reversibility of the modified LDHs as a viable method for capturing CO₂ from the ambient atmosphere (Chávez-Rocha et al. 2023).

The use of surfactants in LDH-MEA samples reduces their adsorption ability, especially with increasing amounts of sodium dodecyl sulfate (SDS). At 80 °C, the presence of surfactants, especially with increasing SDS amounts, leads to a reduction in adsorption ability by 35 to 50%. However, after the MEA extraction process, there is a significant enhancement in CO₂ adsorption, particularly at 55 °C, which could be attributed to the higher concentration of amines present in the samples. The extraction process helps remove surfactants, releasing amino groups for CO₂ adsorption. At a temperature of 80 °C, similar trends are seen, suggesting a substantial increase of 10–30% in the capacity of LDH-MEAs to adsorb CO₂ (Zahedi et al. 2022).

González-Rivera et al. (2022) assessed the thermal stability of LDHs modified by amines by using TGA. The specimens underwent disintegration at three specific temperature ranges: below 150 °C, between 150 and 750 °C, and over 750 °C, respectively. When compared to LDH-MEA, the second step of weight loss exhibited inconsistencies, possibly due to a non-uniform bonding structure that caused several stages

Table 3 The impact of alkali impregnation on various adsorbents under diverse thermal and pressure conditions for CO₂ adsorption

Alkali	W _{impregnating}	LDH	Capacity		Temperature °C	Pressure	References
			Before	After			
K ₂ CO ₃	15%	Mg ₂ Al-CO ₃	0.26	0.52	300	-	Bhatta et al. (2015)
	20%	Mg ₂ Al-CO ₃	0.94	1.43	300	-	Silva et al. (2017)
	20%	Mg ₂ Al-CO ₃	0.45	0.67	300	p _{CO₂} =0.15 bar	Faria et al. (2022)
	20%	Mg ₂ Al-CO ₃	0.41	1.19	350	p _{CO₂} =1 atm	Sun et al. (2020)
	20%	Mg ₂ Al-CO ₃	0.52	1.78	350	-	Hanif et al. (2014)
	20%	Mg ₂ Al-CO ₃	0.26	0.77	450	1.13 bar	Yang and Kim (2006)
	0.13%	Mg ₂ (Al _{0.9} Ga _{0.1}) CO ₃	0.531	1.400	200	0.7 atm	Yavuz et al. (2009)
	20%	Mg ₂ (Al _{0.9} Ga _{0.1}) CO ₃		1.23	400	15%CO ₂	Silva et al. (2019)
	20%	MgAl-CO ₃	0.45	0.65	300	1 bar	Bhatta et al. (2016)
	20%	MgAl-CO ₃	0.10	0.76	400	2 bar, p _{CO₂} =0.4	Oliveira et al. (2008)
	35%	MgAl-CO ₃	0.1	0.94	400	1 atm	Lee et al. (2010)
	35.35%	MgAl-CO ₃	0.083 wet	0.76 wet	403	0.4 atm	Oliveira et al. (2008)
	20%	Mg/Al ratio0.6		1.5	400	Up to 24 bar	Boon et al. (2014a)
	22%	Mg _{0.7} Al _{0.3} -(CO ₃)	0.06	0.37	400	1 bar	Walspurger et al. (2008)
	20%	Mg ₃ Al-CO ₃	0.83	1.21	200	1 atm	Wang et al. (2013)
	20%	Mg ₃ Al-CO ₃	0.53	0.85	300	1 atm	Wang et al. (2011)
	20%	Mg ₃ Al-CO ₃	0.42	1.46	400	-	Zhu et al. (2020a)
	20%	Mg ₃ Al-CO ₃	0.53	0.85	400	1 atm	Wang et al. (2011)
	20%	Mg ₃ Al-CO ₃	0.22	0.83	400	1 atm	Zhu et al. (2019b)
	15%	Mg ₃ Al-NO ₃ /GO nanosheets	0.47	0.6	200	1 atm	Wang et al. (2015)
	12.5%	Mg ₃ Al-stearate	1.01	1.93	300	1atm	Li et al. (2013b)
	-	Mg ₃ Al-CO ₃ /AC	0.156	0.194	200	1 atm	Zhu et al. (2015)
	20%	Mg ₄ Al-CO ₃	0.454	0.96	350	-	Lundvall et al. (2017)
	20%	MG70	0.22	0.72	400	1 atm	Zhu et al. (2019b)
	35%	MG70	0.13	0.68	400	1 atm	Lee et al. (2010)
	20%	LiAl ₂ -CO ₃	0.51	0.83	200	1 atm	(Huang et al. 2014)
NaNO ₃	10%	Ni ₃ Al-CO ₃	0.66	0.85	200	1.2 bar	Hanif et al. (2019)
	20%	Mg ₂ Al-CO ₃	0.41	0.95	200	p _{CO₂} =1 atm	Sun et al. (2020)
	20 wt. %	MG70	0.63	0.77	200	1 atm	Kim and Lee (2019)
	23.6 wt. %	Mg ₉ Al - CO ₃		9.27	240	1 atm	Kim et al. (2016a)
	30 wt. %	MgAl-CO ₃	0.46	1.14	200	1 atm	Kim and Lee (2019)
KNO ₃	20%	Mg ₂ Al-CO ₃	0.41	0.83	200	p _{CO₂} =1 atm	Sun et al. (2020)
Na/KNO ₃	20%	Mg ₂ Al-CO ₃	0.41	1.07	200	p _{CO₂} =1 atm	Sun et al. (2020)
Cs ₂ CO ₃	20%	MgAl-CO ₃	0.10	0.44	400	2bar p _{CO₂} =0.4bar	Oliveira et al. (2008)
(Li - Na - K) NO ₃	55 mol%	Mg ₇ Al - CO ₃ - C ₁₆	0.91	3.25	200	p _{CO₂} =1 atm	Qin et al. (2017a)
K	20%	Mg ₂ (Al _{0.9} Ga _{0.1}) CO ₃	0.45	1.82	300	1.08 bar	Miguel et al. (2014)
Na	5%	Mg ₂ Al-CO ₃	0.45	0.65	300	p _{CO₂} =0.15 bar	Faria et al. (2022)
K, Na	18.5% K, 1.5% Na	Mg ₃ Al-CO ₃	0.28	1.109	350	1.34 bar	Martunus et al. (2012)
Cs	10%	Mg ₂ Al-CO ₃	0.45	0.64	300	p _{CO₂} =0.15 bar	Faria et al. (2022)
	20%	Mg ₂ (Al _{0.9} Ga _{0.1}) CO ₃	0.45	1.4	300	1.08 bar	Miguel et al. (2014)

of dehydroxylation. The decrease in weight seen at temperatures below 150 °C may be due to the removal of interstitial water molecules. The weight was significantly reduced at temperatures over 750 °C as a result of the decomposition of sulfate species residue (González-Rivera et al. 2022).

The weight loss in the second and third stages was lower on the ultrasonic route (UH) compared to the UB route (54–66%), indicating that the UH route has the capability to provide a more stable material in comparison to conventional procedures. Following the amine modification, the

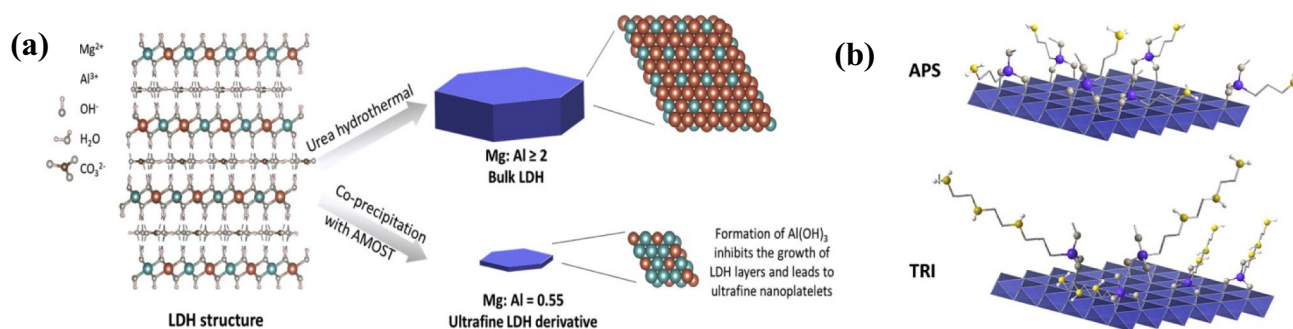


Fig. 16 **a** Bulk and ultrafine amine monolayer-grafted MgAl-LDHs synthesis, **b** grafted amine monolayer structure (Chávez-Rocha et al. 2023)

degradation patterns of LDHs treated with MEA showed significant differences in comparison to untreated LDHs produced using various methods. Samples created using sonochemistry, such as UB-MEA and UH-MEA, demonstrated improved thermal stability in comparison to traditional samples. This implies that using ultrasonic technology may enhance the uniform distribution of surfactant in LDH generation, easing the extraction of surfactant during MEA extraction and enhancing the stability of the substance (González-Rivera et al. 2022).

CO₂ capture using improved hydrotalcite

Hydrotalcite-based sorbents, a class of anionic clays, have emerged as promising candidates for high-temperature CO₂ capture due to their favorable properties, including lower regeneration energy requirements, good CO₂ adsorption kinetics, and stability over multiple cycles (Yong and Rodrigues 2002; Zhu et al. 2020a; Faria et al. 2022). These properties make them suitable for effective and efficient CO₂ capture in various industrial applications (Faria et al. 2022). Hydrotalcite is influenced by factors such as the pH during synthesis, molar ratios of its constituent elements, and strategies to increase its surface area (Faria et al. 2022; Suescum-Morales et al. 2022; Trujillano et al. 2023; Permanasari et al. 2024). It also notes that their adsorptive capabilities are substantially influenced by the aluminum content, the types of anions included, and the level of hydration (Yong et al. 2001; Zhang et al. 2020a). The role of environmental conditions such as temperature, pressure, and humidity is also crucial (Faria et al. 2022; Suescum-Morales et al. 2022). Hydrotalcite CO₂ capture capacity has traditionally been lower than other high-temperature sorbents, prompting research into improving their performance through various synthetic modifications (Kim and Lee 2019; Liu et al. 2024). A significant enhancement in CO₂ uptake is achieved through calcination, which alters the structure of hydrotalcite. This process imparts a “memory effect” to calcined hydrotalcites

(CLDHs), allowing them to return to their original structure in humid and CO₂-rich environments, thus boosting CO₂ capture and storage (Kim and Lee 2019; Suescum-Morales et al. 2022).

The study conducted by Yong et al. (2001) investigates the effectiveness of hydrotalcite-like compounds (HTLcs) in capturing carbon dioxide at elevated temperatures, focusing on various commercially available HTLcs differentiated by their aluminum content, types of anions, and water content. The research highlights that optimal aluminum content and specific heat treatment temperatures are crucial for enhancing CO₂ adsorption capacities. It was found that carbonate anions, due to their size and charge, are more effective than hydroxide anions in CO₂ adsorption, as they expand the interlayer spacing of HTLcs, thereby boosting adsorption capacity. Additionally, water presence was found to diminish adsorption capacity, while a lower aluminum content tends to increase it, which is significant for refining HTlc design for high-temperature CO₂ capture from exhaust gases (Yong et al. 2001). Similarly, the study by Soares et al. (2005) emphasizes how structural factors such as aluminum content and carbonate anions enhance interlayer spacing, improving adsorption capacity. Through dynamic and static adsorption tests, the study demonstrates HTLcs efficiency in handling variations in CO₂ concentrations and gas flow rates. Despite moisture reducing efficiency, HTLcs exhibit strong thermal regeneration capabilities, allowing for multiple adsorption–desorption cycles without significant performance loss, making them a cost-effective solution for continuous high-temperature CO₂ capture in industrial settings (Soares et al. 2005).

The study by Moreira et al. (2006) supports the effectiveness of HTLcs for high-temperature CO₂ capture, aligning with the findings that hydrotalcite-based sorbents exhibit favorable properties such as good CO₂ adsorption capacity, stability over multiple cycles, and lower regeneration energy requirements. It highlights the ability of HTLcs to maintain strong adsorption capacity across a temperature range of

29 to 350 °C, emphasizing their suitability for industrial CO₂ capture applications, particularly in flue gas treatment. Additionally, the study confirms the thermal regeneration capability of HTlcs, which can undergo multiple adsorption–desorption cycles without significant performance loss, provided that the regeneration temperature is maintained above 370 °C. This complements other findings on the memory effect of CLDHs (Kim and Lee 2019; Suescum-Morales et al. 2022). Moreira et al. (2006) also identified that HTlcs adsorption efficiency remains consistent over time, making them a viable, cost-effective solution for continuous industrial CO₂ capture processes (Moreira et al. 2006).

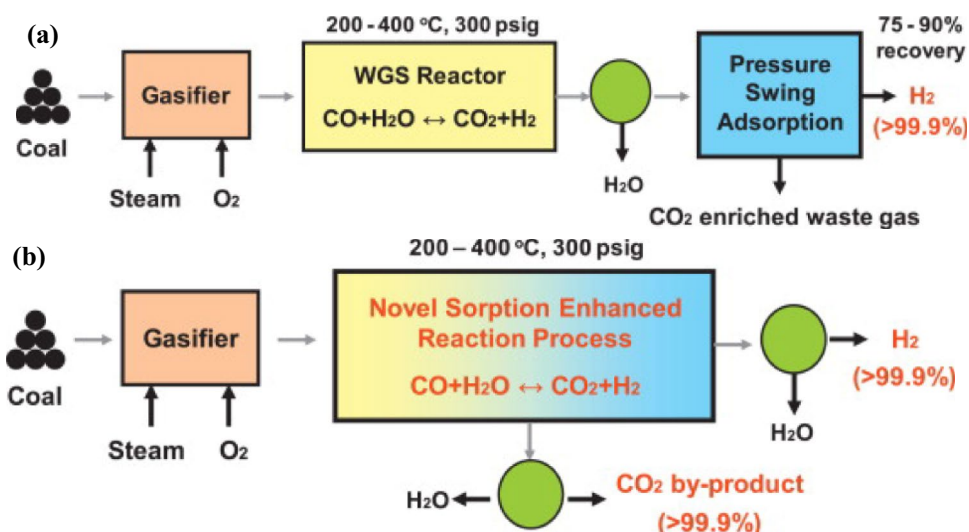
A study by Hanif et al. (2014) explored several synthetic routes to enhance the CO₂ capture capacity of Mg–Al hydrotalcites. The researchers employed three primary methods: conventional co-precipitation, ultrasound-assisted co-precipitation, and microwave irradiation of the precipitating gel. These techniques were designed to disrupt the typical layer stacking of hydrotalcites, leading to an increase in surface area and a reduction in crystallite size. The study found that both ultrasound and microwave-assisted methods significantly enhanced the surface area by two to five times compared to the conventional method, which correlated with improved equilibrium and dynamic CO₂ sorption capacities (Hanif et al. 2014). Additionally, the study investigated the promotional effect of K₂CO₃ impregnation on hydrotalcites. K₂CO₃, known for increasing the basicity of hydrotalcites, was found to enhance CO₂ capture capacities further, particularly at higher temperatures. The extent of this enhancement was temperature-dependent, with CO₂ capacity improvements ranging from 34% to over 100% depending on the synthesis method and operating temperature. For instance, the microwave-aged hydrotalcite impregnated with K₂CO₃ exhibited the highest CO₂ capacity at 10 bar, surpassing many reported values in the literature for similar materials. The findings from this study suggest that with proper synthetic modifications, hydrotalcite-based sorbents can become highly effective for high-temperature CO₂ capture. The enhancements achieved through microwave and ultrasound synthesis, along with K₂CO₃ promotion, indicate that these materials can meet the demanding requirements of industrial CO₂ capture processes, offering a viable alternative to traditional methods. This research contributes to the broader field of CO₂ capture by providing insights into material design strategies that optimize performance under specific operational conditions (Hanif et al. 2014). A study by Wu et al. (2013b) shows that promoting HTlcs with KNO₃ significantly improves CO₂ adsorption, particularly in high-temperature settings, similar to K₂CO₃ modifications. These chemical enhancements boost the basicity and reactive surface area of the sorbents, enabling greater CO₂ capture under challenging industrial conditions. This demonstrates the importance of chemical promotion in optimizing

hydrotalcite sorbents for effective and sustainable CO₂ capture technologies (Wu et al. 2013b).

The study by Kim et al. (2016a) investigates the impact of varying Mg/Al molar ratios on the CO₂ sorption capacities of hydrotalcites, synthesized through a modified coprecipitation method for high-temperature applications like direct CO₂ capture from flue gases. It explores ratios ranging from 3 to 30, discovering that a well-crystallized hydrotalcite structure is formed at a ratio of 3. However, as the ratio increases, the presence of both crystalline hydrotalcite and unreacted Mg(OH)₂ phases emerges due to excess magnesium from the precursor, which affects the crystallinity as evidenced by broader peaks in XRD analysis (Kim et al. 2016b). A significant enhancement in CO₂ sorption capacity is noted with the presence of NaNO₃, formed due to excess magnesium nitrate used in the synthesis, achieving a peak uptake of 9.27 mol/kg at 240 °C and 1 atm CO₂. This optimal NaNO₃ concentration significantly boosts CO₂ capture but, when excessive, can lead to pore blockage, reducing the surface area and, consequently, the sorption capacity. The study also reveals unique two-step CO₂ sorption behavior in hydrotalcites with higher Mg/Al ratios initially low uptake followed by a sharp increase. This behavior suggests a complex interplay between physical adsorption and chemical reactions within the material, particularly the formation of MgCO₃ from the reaction of MgO and CO₂. These insights highlight the importance of optimizing the Mg/Al ratio and NaNO₃ content to maximize the efficiency of CO₂ capture in high-temperature environments (Kim et al. 2016a).

The study of Bhatta et al. (2015) explores advancements in CO₂ capture technology, focusing on the enhancement of adsorption capacities at high temperatures using HTlcs. These enhancements are achieved through the incorporation of a novel, cost-effective support material derived from coal, known as coal-derived graphitic material (CGM), and further improved by impregnating these supported HTlcs with K₂CO₃. The research demonstrates that using CGM as a support material significantly increases the CO₂ adsorption capacity of the HTlcs, recording a fresh adsorption capacity of 1.10 mmol g^{−1} at 300 °C. This capacity is maintained over multiple cycles, with an average working capacity of 0.42 mmol g^{−1} observed during nine cycles of adsorption and desorption between 300 and 400 °C. This performance enhancement is attributed to the increased surface area and density of active sites provided by the CGM. The K₂CO₃ impregnation plays a crucial role in this process by increasing the number of active sites despite a reduction in the overall surface area and pore volume, which typically would hinder adsorption capacity. This combination of CGM support and K₂CO₃ impregnation not only improves the efficiency of CO₂ capture but also suggests a viable method for scaling up this technology for industrial applications, particularly in the treatment of flue gases. The study utilizes the Yoon-Nelson

Fig. 17 **a** Standard process diagram for hydrogen generation through coal gasification. **b** Innovative sorption-enhanced reaction technique for hydrogen production via coal gasification (Jang et al. 2012)



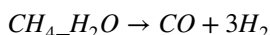
kinetic model to accurately fit the experimental adsorption data, indicating the model applicability in designing and optimizing the adsorption process. The approach outlined in this research provides a promising direction for enhancing the economic and operational feasibility of high-temperature CO₂ capture technologies (Bhatta et al. 2015).

CO₂ capture within the framework of steam methane reforming

The sorption enhanced reaction process (SERP) represents an advanced approach in the field of steam methane reforming (SMR), designed to simultaneously enhance hydrogen production and CO₂ capture (Lim et al. 2023b). The SERP utilized in SMR is strategically divided into two key reaction steps to optimize both hydrogen production and CO₂ capture:

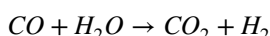
1. Steam methane reforming reaction

Methane (CH₄) reacts with steam (H₂O) in the presence of a catalyst, typically at high temperatures, to produce hydrogen (H₂) and carbon monoxide (CO). This reaction is represented by the following equation:



2. Water gas shift reaction

In the subsequent phase, CO produced in the first step reacts with more steam to produce additional H₂ and CO₂. This reaction is facilitated by a catalyst and is crucial for increasing hydrogen yield while producing CO₂ as a byproduct, which can be captured in situ (Jang et al. 2012). The reaction equation is:



Illustrated by Fig. 17, the process showcases innovative setups where SERP operates between 200–400 °C and at pressures around 300 psig to optimize CO₂ capture while maintaining high hydrogen yield. These figures provide a visual representation of the operational flow and efficacy of CO₂ capture, highlighting how SERP not only enhances the purity and yield of hydrogen but also significantly mitigates CO₂ emissions.

The integration of sorption into this reaction process, as embodied in SERP, aims to optimize both the reforming efficiency and CO₂ capture. SERP utilizes a sorbent material that selectively adsorbs CO₂ from the reforming gas stream, thereby shifting the chemical equilibrium towards increased hydrogen production. This shift occurs because the removal of CO₂ from the reaction mixture favors the formation of additional hydrogen (Cañada-Barcala et al. 2023). The operational efficacy of SERP is significantly influenced by the choice of sorbent material (Beaver et al. 2009). Effective sorbents must exhibit high CO₂ affinity, optimal adsorption/desorption kinetics, and durability under the high temperatures typical of reforming conditions. Common sorbents include calcium oxide (CaO), lithium-based compounds, and various metal oxides, each selected for their ability to capture CO₂ efficiently. The sorbent material can be integrated directly within the reforming reactor or utilized in a separate unit designed for CO₂ capture. The successful application of SERP necessitates not only effective CO₂ capture but also efficient regeneration of the sorbent. Regeneration involves heating the sorbent to release the captured CO₂, which is then collected for storage or further use. Ensuring that the sorbent retains its capacity and mechanical integrity over multiple cycles of adsorption and regeneration is crucial for maintaining process efficiency (Sun et al. 2023).

By integrating sorption directly into the reforming process, SERPs improve efficiency and reduce emissions

(Fatigati et al. 2021; Lee et al. 2022). A specific application of this technique is the sorption-enhanced water gas shift (SEWGS) reaction using K_2CO_3 -promoted hydrotalcite (Jang et al. 2012). The SEWGS process uniquely combines the water gas shift reaction with the selective chemisorption of CO_2 , leading to the direct production of nearly pure hydrogen (Malsegna et al. 2024). This integration effectively overcomes the thermodynamic limitations of traditional water gas shift reactions by removing CO_2 from the reaction zone, thereby enhancing the reaction rate and the overall yield of hydrogen (Lee et al. 2007). The article of Jang et al. (2012) highlights the use of K_2CO_3 -promoted hydrotalcite as a CO_2 chemisorbent, which is critical in continuously removing CO_2 and driving the reaction towards greater hydrogen production. It underscores that using K_2CO_3 -promoted hydrotalcite in the SEWGS process offers a viable pathway for efficient high-purity hydrogen production while enabling significant CO_2 capture, thereby contributing to more sustainable industrial practices (Jang et al. 2012). Experimental and simulation studies analyze the impact of various operational parameters such as temperature, pressure, and H_2O/CO feed ratio on the process efficiency. Results indicate that higher water-to- CO ratios and lower temperatures generally favor hydrogen productivity and CO conversion, whereas higher pressures increase hydrogen output but can decrease CO conversion (Lee et al. 2007; Jang et al. 2012).

Wu et al. (2013a) demonstrated that K_2CO_3 -promoted hydrotalcites exhibit high CO_2 capture capacities, with the highest reported being 1.13 mol CO_2 per kg at 656 K. The K -promoted hydrotalcites also showed excellent cyclic stability, retaining over 93% of their sorption capacity after 10 adsorption/desorption cycles. The regeneration of sorbents can be performed through temperature swing (TSA) operations, significantly reducing the time required for desorption (Wu et al. 2013b).

Recent advancements include the development of Ni-substituted hydrotalcites, which combine CO_2 adsorption and methanation capabilities into a single material. In a study by Catarina Faria et al. (2023), Ni-hydrotalcites promoted with cesium were shown to capture approximately 0.24 mol CO_2 per kg at 350 °C and 1 bar, followed by complete conversion of the captured CO_2 into methane with high selectivity ($\geq 98.1\%$) through the Sabatier reaction. Two cesium impregnation routes, powder impregnation (PI) and extrudate impregnation (EI), were tested, revealing that Cs promotion significantly improved both CO_2 capture and methanation (Catarina Faria et al. 2023). A follow-up study further optimized these materials by examining the effects of temperature, pressure, and flow rates on the performance of the DFMs. It was found that increasing pressure enhanced both CO_2 capture and methane purity, while higher H_2 flow rates accelerated the methanation process. The materials exhibited long-term

stability, maintaining high performance for up to 250 h of cyclic operation, with an impressive 100% CH_4 selectivity under optimal conditions (Faria et al. 2023). These dual-function materials enhance SERP efficiency by capturing CO_2 near catalytic sites, reducing resistance to heat and mass transfer, and eliminating the need for separate reactor systems, thus intensifying the overall process (Catarina Faria et al. 2023).

Martins et al. (2019) explore the use of NiMgAl, Ru/NiMgAl, and Ru/SiO₂ catalysts for CO_2 methanation, focusing on converting CO_2 into methane through the Sabatier reaction. Among these, NiMgAl and Ru/NiMgAl demonstrated the highest performance, achieving up to 90% CO_2 conversion and near-complete methane selectivity at 300 °C. These catalysts also showed good stability, making them suitable for long-term use. NiMgAl was particularly promising due to its lower production cost and consistent performance, though some deactivation occurred at 350 °C due to carbon deposition blocking active sites. In contrast, Ru/SiO₂ had the lowest performance, suffering significant deactivation at higher temperatures. Overall, the hydrotalcite-derived catalysts, especially NiMgAl and Ru/NiMgAl, offer great potential for power-to-methane (PtM) technologies by integrating efficient CO_2 capture and methanation (Martins et al. 2019).

In conclusion, the SERP is a key advancement in SMR, enabling efficient hydrogen production and CO_2 capture. Materials like K_2CO_3 -promoted and Ni-substituted hydrotalcites enhance adsorption, methanation, and regeneration, improving both efficiency and sustainability. SERP not only boosts hydrogen yields and reduces CO_2 emissions but also has the potential to lower costs by minimizing the need for separate CO_2 capture units. However, challenges remain in optimizing sorbents and reactor design. Continued research into materials and process control will maximize SERP potential, making it a vital pathway for sustainable energy and carbon reduction, especially when integrated with PtM technologies (Cañada-Barcala et al. 2023; Udemu and Font-Palma 2023; Faria et al. 2023).

Effect of the synthesis method of LDHs-based materials on CO_2 capture

The synthesis method of LDHs significantly impacts CO_2 capture efficiency. Different methods like co-precipitation and urea hydrolysis yield LDHs with varying properties. LDHs synthesized via co-precipitation, the most common method, show high yields, purity, and high CO_2 adsorption capacities. However, this method may result in less uniform particle sizes and lower surface areas, limiting the exposure of active sites necessary for effective CO_2 adsorption. On the other hand, LDHs synthesized via the

urea hydrolysis method exhibit better cyclic stability (Ge et al. 2023). Hydrothermal synthesis produces LDHs with higher crystallinity and uniformity, enhancing stability and active site density for improved CO₂ adsorption, albeit with higher energy consumption and longer synthesis times (Kameliya et al. 2023). Sol–gel processes introduce organic components to boost porosity and surface area, aiding CO₂ diffusion rates and capture efficiency, but require precise chemical control to prevent structural collapse (Maegawa et al. 2023). Each method offers distinct advantages and challenges, influencing the practicality and efficiency of LDHs in CO₂ capture applications. Additionally, the choice of metal salt precursors influences the physical properties of LDHs and the CO₂ capture performance of derived materials (Fan et al. 2023). Furthermore, the synthesis conditions and metal ratios affect the crystallite phase formation and redox reactivity of LDH-derived materials, impacting their CO₂ adsorption capacities and stability (Ibrahimova 2022).

The study highlights the impact of various factors on the CO₂ adsorption capabilities of LDHs. Initially, the addition of surfactants reduces adsorption capacity because surfactant anions alter the charge of amino groups, with significant effects observed particularly when larger quantities of SDS are introduced and at elevated temperatures around 80 °C, though this effect diminishes by 35–50% subsequently. The removal of MEA enhances CO₂ adsorption in LDH-MEA samples, especially notable at 55 °C, attributed to increased amine concentrations following surfactant extraction which frees up amino groups for CO₂ adsorption. This trend is also evident at 80 °C, showing a 10–30% increase in adsorption capacity (Hou et al. 2022).

Additionally, the study examines the influence of sonication on CO₂ adsorption. Traditional precipitation methods generally perform better than sonicated samples at 55 and 80 °C due to sonication facilitating intense chemical reactions and reducing the availability of active sites for CO₂ adsorption. Controlled sonication, however, particularly at milder intensities, improves CO₂ adsorption by optimizing the molecular structure and increasing the availability of amino groups for effective CO₂ capture (Shokrollahi et al. 2023).

Overall, the research indicates that both the chemical manipulation of LDHs, such as surfactant removal and amine enhancement, and the physical processing methods like sonication, significantly affect their CO₂ adsorption efficiency. These findings suggest tailored approaches in adsorbent preparation could enhance CO₂ capture performance, emphasizing the role of controlled sonication and chemical treatment in optimizing LDH-based adsorbents (González-Rivera et al. 2022). These findings underscore the importance of the synthesis method in tailoring LDHs for efficient CO₂ capture applications.

Comparison of CO₂ capture capabilities of different LDH-based materials

LDHs are a class of materials that offer considerable versatility due to their unique lamellar structure and tunable chemical composition. The modification and functionalization of LDHs have emerged as critical strategies to enhance their performance across various applications, including catalysis, adsorption, drug delivery, and environmental remediation. These modifications significantly alter the physical, chemical, and structural properties of LDHs, enabling the design of materials with tailored functionalities (Li et al. 2024a). One of the primary modification strategies involves morphological control, which focuses on altering the size and shape of LDH particles. By adjusting these parameters, researchers can optimize the surface area and pore structure, thereby enhancing the reactivity and adsorption capacity of the material. For example, reducing particle size to the nanoscale increases the surface-to-volume ratio, providing more active sites for reactions or adsorption processes. Additionally, the creation of specific morphologies, such as nanosheets or nanoplates, can expose more reactive facets, improving the material overall effectiveness in various applications (Wei et al. 2022b).

Alkali metal doping is another significant modification technique that influences the electrochemical properties of LDHs. By introducing alkali metals such as sodium or potassium into the LDH structure, the material basicity and ion-exchange capacity can be significantly enhanced. This doping process increases the positive charge density within the layers, which strengthens the electrostatic interactions with anions, thereby improving the material ability to adsorb target species. This modification is particularly beneficial in catalytic processes and environmental remediation, where strong basic sites are essential for optimal performance (Yao et al. 2024). Compositional alteration, involving the substitution of metal cations within the LDH layers, further expands the functionality of these materials. By replacing common cations like magnesium or aluminum with transition metals such as iron or cobalt, the electronic and thermal properties of LDHs can be finely tuned. This compositional flexibility allows for the introduction of redox-active sites, which are critical for catalysis and other applications requiring electronic conductivity or enhanced thermal stability (Liu et al. 2024).

Intercalation of organic molecules between the LDH layers represents a powerful functionalization approach. This process not only stabilizes the LDH structure but also introduces new chemical functionalities, such as increased hydrophobicity or enhanced catalytic activity. The intercalated organic molecules can participate directly in chemical reactions or act as stabilizing agents, making LDHs more effective in applications like drug delivery and heterogeneous catalysis. Furthermore, organic intercalation can prevent the collapse of LDH layers,

ensuring the material long-term structural integrity under various operating conditions (Mohammadi et al. 2022). Surface functionalization is another key strategy for enhancing the performance of LDHs. By grafting functional groups, such as hydroxyl, carboxyl, or amino groups, onto the surface of LDHs, the material interaction with specific molecules can be significantly improved. These functional groups can form strong bonds with target compounds, increasing the material selectivity and efficiency in adsorption and catalysis. Additionally, surface functionalization can improve the dispersion of LDHs in various media, which is crucial for applications in composite materials and catalytic processes (Pustahija and Kern 2023).

Hybridization of LDHs with other materials, such as carbon-based compounds or metal oxides, further broadens their application potential. The creation of LDH-based composites combines the intrinsic properties of LDHs with the unique characteristics of the hybridized material. For instance, hybridizing LDHs with graphene oxide can enhance their electrical conductivity and mechanical strength, making these composites ideal for use in energy storage devices and electrochemical catalysts (Liu et al. 2024). Thermal treatment, including calcination, is a widely used modification method that transforms LDHs into mixed metal oxides. This process significantly enhances the surface area, porosity, and catalytic activity of the material by decomposing the hydroxide layers and exposing more active sites. Calcined LDHs are particularly effective in high-temperature applications, such as catalytic processes and CO₂ capture. Moreover, the rehydration of calcined LDHs can restore the original layered structure with modified properties, allowing for the material reuse in cyclic applications (Yuliasari et al. 2023).

The anion-exchange capacity of LDHs can be exploited through the exchange of interlayer anions with functional anions. This modification enables the tailoring of the material chemical reactivity, stability, and adsorption capacity. Anion exchange is especially valuable in applications where the selective capture of specific anions is required or where the interlayer spacing needs to be adjusted to accommodate larger guest molecules (Tang et al. 2022). The modification and functionalization of LDHs are critical for enhancing their properties and expanding their applicability in various fields. Through careful selection of modification strategies, researchers can design LDH-based materials with optimized performance, making them highly effective for a wide range of industrial, environmental, and biomedical applications (Oosthuizen et al. 2023).

Effects of process conditions on CO₂ capture using LDH-based materials

The process parameters have a significant impact on CO₂ capture using LDH-based materials (Belskaya et al. 2023). Subject to a range of external factors, including

temperature, pressure, humidity, and steam. The efficiency of CO₂ adsorption using LDHs decreases as temperatures rise, mainly because the adsorption process is exothermic. However, greater temperatures can be beneficial during the desorption phase for regenerating the material. Elevated pressure increases CO₂ absorption efficiency by increasing the gas partial pressure, enhancing its interaction with LDHs. The presence of humidity can hinder the process of CO₂ adsorption by competing for the active sites on the surface of LDH. However, it may also facilitate the opening of the material structure, improving the intercalation of gas. Steam is commonly employed throughout the regeneration process to aid in the elimination of CO₂ and prepare LDHs for further use. Comprehending and controlling these ecological parameters is crucial for maximizing the efficiency of LDHs in CO₂ capture applications (Cheah et al. 2022). This section will comprehensively examine how the specified parameters influence the capture of CO₂ using LDHs.

Effect of temperature

Temperature affects the adsorption of CO₂ by LDH-based materials, but adding additives and modifying the material can lessen this impact (Donat and Müller 2022). This process improves cycle stability and mechanical strength, making it suitable for industrial applications. For instance, the addition of CaO to sorbents based on hydrotalcite improves their capacity to collect CO₂ at elevated temperatures. Adding 12-aminododecanoic acid to LDH layers as an intercalator makes CO₂ separation work better by extending the distance between the layers and making gas separation better. LDHs intercalated with adamantanecarboxylate are very good at absorbing CO₂ and cycling it at high temperatures (Manohara et al. 2020).

This makes them very promising candidates for making better sorbents that can capture CO₂ efficiently in a wide range of situations. Modified synthesis procedures, including amide hydrolysis and pelletization, may be used to create structured sorbents that improve the capacity to collect CO₂ at ambient and medium temperatures. These techniques also increase cycle stability and mechanical strength, making them suitable for industrial applications. The research on MgAl-MMO indicates that the adsorption capacity of calcined LDH diminishes as the operating temperature increases, which is consistent with the changes in composites and the conversion of M²⁺ to lithium. Impregnating with potassium carbonate enhances thermal stability, enabling increased working temperatures in pre-combustion settings and maintaining comparable adsorption capacities throughout the temperature range of 200–500 °C. Experiments involving magnesium and lithium demonstrate alterations in their behavior, regardless of the presence or absence of K₂CO₃ (Ai et al. 2023).

The adamantanecarboxylate-intercalated CaAl-LDH exhibits a peak at 650 °C during the adsorption process, whereas regeneration necessitates temperatures above 900 °C. The large surface area of the organic–inorganic composite and the presence of MMOs on a substantial carbon sheet make it a possible CO₂ absorbent at high temperatures. The inclusion of amorphous aluminum-containing phases in metal–organic frameworks (MOFs) acts as a surface for the active sorbent calcium oxide (CaO), thereby improving the diffusion and efficiency of CO₂ capture. A TGA was performed in the temperature range of 200–700 °C using a gas combination consisting of 86% CO₂ and 14% N₂, with a flow rate of 100 mL/min. The results demonstrated a gradual increase in the ability of CO₂ to be adsorbed as the temperature rose from 200 to 600 °C. The CO₂ capture capabilities were determined at 5.18 and 4.9 mmol/g at temperatures of 650 and 700 °C, respectively. At a temperature of 600 °C, the carbonation process effectively transformed almost all of the calcium present in the nanocomposite, accounting for about 27% of its weight, or 6.13 mmol/g, into carbonate. The nanocomposite sorbent underwent regeneration in a nitrogen environment at temperatures of 700, 800, and 900 °C (Manohara et al. 2020). The nanocomposite was subjected to a series of 60 cycles at temperatures ranging from 600 to 700 °C, with a CO₂ concentration of 86%, in order to evaluate its performance in high-temperature CO₂ cycling. The sample demonstrated exceptional cycle stability, achieving almost 100% sorbent regeneration during a 30-min regeneration time. The nanocomposite carbonation process happens rapidly, with about 90% of carbonation occurring during the first 5 min of the cycle. Even after 60 cycles, the nanocomposite maintained its initial structure and porosity (Zhang et al. 2023a).

Effect of pressure

The influence of pressure on CO₂ collection using LDH-based materials can be reduced by integrating different additives and implementing adjustments. Including 12-aminododecanoic acid inside the layers of MgAl LDH enhances the efficiency for CO₂ adsorption. This enhances the effectiveness of separation by increasing the distance between the layers and making use of the quadrupole moment effects (Zainuddin et al. 2023).

Amide hydrolysis and aqueous exfoliation techniques have been used to boost CO₂ capture capacity and create structured LDH-based MMOs with superior properties, including increased capture capability at different temperatures and greater mechanical stability. It is essential to comprehend the adsorption capabilities of MMO samples under extreme pressure. Although there is increasing interest in CO₂ adsorption at moderate temperatures, some

operating circumstances, such as steam and high pressure, have been postponed due to displacement effects. Operational requirements often require pressures of around 10 bar for processes such as water gas shift or steam reforming of methane (Cheah et al. 2022). The study conducted by El Batouti et al. (2023) demonstrates that their samples exhibit a two-fold increase in adsorption capacity when subjected to microwave radiation applied to the gel generated by co-precipitation (El Batouti et al. 2023). Similarly, research conducted in Germany using the Sasol commercially available KMG-30 product also demonstrated a similar enhancement in adsorption capacity when microwave-assisted methods were employed for hydrogel synthesis (da Silva et al. 2020).

Boon et al. (2014a) constructed a SEWGS reactor model in order to examine the competitive adsorption of CO₂ and H₂O. The CO₂ adsorption process of KMG-30 can be divided into two stages: the first stage involves interaction with surface sites, while the second stage concentrates on adsorption within nanopores. A number of cobalt-magnesium–aluminum LDH compounds were made with different amounts of M²⁺ ions. The compounds exhibited a volumetric capacity of about 35% more than the reference material when subjected to a temperature of 375 °C and a pressure of 21 bar (Boon et al. 2014a).

A commercially accessible LDH was modified by adding 5 wt.% cobalt, iron, or nickel. The researchers evaluated the adsorption capacities of these modified LDHs under conditions of temperatures reaching 300 °C and pressures up to 44 bar. The sample treated with nickel exhibited a greater adsorption capacity in comparison to the other samples, most likely because chemisorption takes place at robust and compact basic sites (Ai et al. 2023).

Ramírez-Moreno et al. (2014) emphasize the significance of investigating high-pressure settings. The researchers conducted a comparison of the adsorption properties of MgAl LDH, which was synthesized using the coprecipitation approach with an M²⁺/M³⁺ ratio of 3:1 at several temperatures and pressures with low supersaturation (Ramírez-Moreno et al. 2014). In Fig. 18, the graph illustrates the relationship between the breakthrough capabilities of pure CO₂ and the partial pressure of CO₂. The breakthrough capacities are represented by diamonds, while the expected capacity is shown by a line. The contribution of different locations is indicated by dashed lines.

Effect of moisture

The presence of water vapor can affect CO₂ adsorption in both pre-combustion and post-combustion processes. The efficacy of trapping CO₂ using LDHs-based materials is greatly affected by water vapor (Ding et al. 2023). Studies indicate that the presence of water vapor enhances the

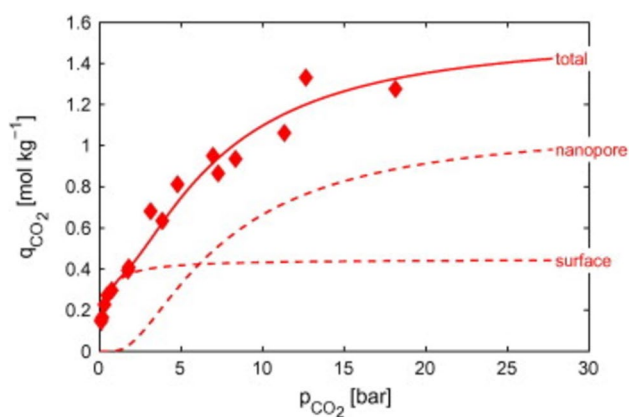


Fig. 18 The graph illustrates the relationship between pure CO₂ breakthrough capabilities and partial pressure (Boon et al. 2014b)

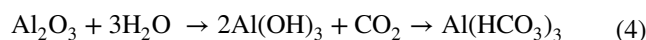
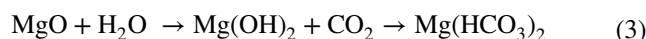
effectiveness of MMOs in adsorbing CO₂ by improving both the adsorption and desorption mechanisms. Water can react with metal oxides, resulting in the formation of hydroxides. These hydroxides then react with CO₂ to produce bicarbonates (Huang 2023).

MMOs had the ability to partly repair LDH structure in dry settings and completely repair interlayer gap in wet situations by generating bicarbonate species. Nevertheless, a high water vapor content might adversely impact the situation by diminishing the size of pores and intensifying the resistance to the spread of adsorbed CO₂. The XRD diffractograms obtained at 10 bar revealed the rapid production of potassium carbonate while transitioning from dry to wet conditions. Furthermore, magnesium carbonate slowly arose and metal oxides disappeared (Luong et al. 2023).

Iruretagoyena et al. (2023) did a study using TGA and a packed bed reactor to look into the factors that affect how CO₂ and H₂O stick to commercial hydrotalcite (KMG30, Sasol) at different temperatures and chemical makes-ups. The researchers identified four distinct adsorption sites, one specifically for H₂O and another specifically for CO₂, both possessing the characteristic of quick replenishment. The first category of sites diminishes as the temperature rises, but the second category intensifies. The last two categories of sites integrate both components and need a greater amount of exertion to restore. Utilizing steam in the desorption phase of the cycle provides many benefits, such as expedited and enhanced elimination of the adsorbate and superior performance in subsequent cycles. Steam increases methane generation, which helps to release CO₂ more quickly and efficiently compared to dry conditions. This, in turn, improves productivity in the subsequent cycle. Water may act as the first stage in the adsorption process (Iruretagoyena et al. 2023). In order to preserve interlayer water, Karthikeyan et al. (2021) subjected exfoliated NiAl-LDH to a 30-min steam pre-treatment, resulting in a material that is 85% effective in adsorbing CO₂ (Karthikeyan

et al. 2021). LDH/CMC films enhance their gas barrier characteristics in humid environments as a result of the formation of hydrogen bonds with water molecules. LDHs, when intercalated with adamantanecarboxylate, exhibit exceptional CO₂ capture efficiency under both high-temperature and industrial flue gas conditions (Zhao et al. 2023b).

Ram Reddy et al. (2008) studied the effect of water present in the feed. The study presented an evidence that water in the feed does not alter the CO₂ sorption rate by LDOs, but it significantly improves their efficiency in capturing CO₂. In experiments conducted with a moist gas stream, the efficiency of CO₂ capture by LDOs increased from 0.61 mmol/g under dry conditions to 0.71 mmol/g. This increase in efficiency is linked to the interaction between water and the LDO, leading to the production of magnesium and aluminum hydroxides. These hydroxides then react with CO₂, forming bicarbonates, as shown in the equations, and thereby enhancing the efficiency of CO₂ capture without altering the overall sorption capacity (Ram Reddy et al. 2008). Figure 19 illustrates the alterations in the sorption performance of LDO (low-density oxygen) in the transition from a wet to a dry CO₂ environment.

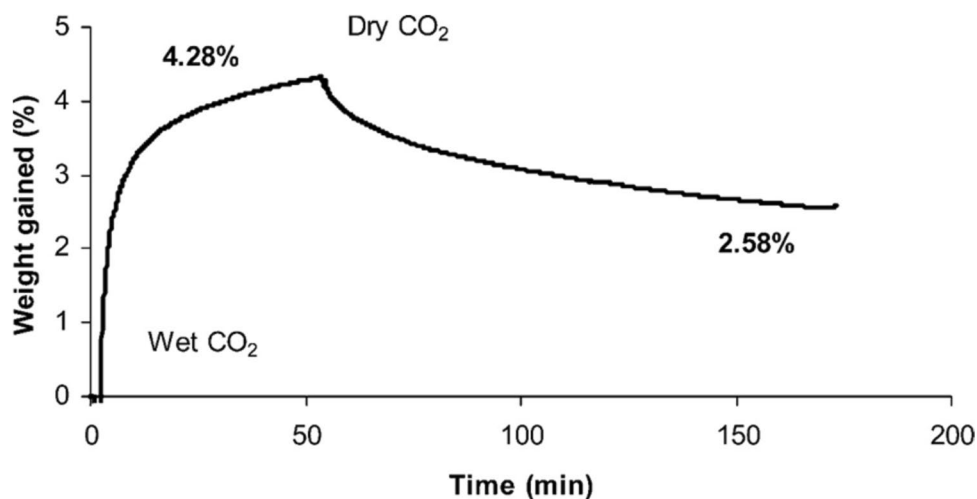


Effect of gas composition

The effect of gas composition on the capabilities of LDHs in adsorbing CO₂ is significant, influencing several key aspects of its performance. Firstly, the concentration of CO₂ in the gas mixture directly impacts the sorption capacity and efficiency of LDH, with higher concentrations generally enhancing CO₂ interaction and uptake. Studies have shown that LDHs, such as CO₃·Mg–Al LDHs, exhibit high CO₂ adsorption capacities even at low CO₂ concentrations (Kameda et al. 2023). The presence of other gases like nitrogen or oxygen can indeed impact the selectivity and efficiency of CO₂ adsorption in LDHs by competing for adsorption sites (Fan et al. 2023; Ai et al. 2023). The chemical interactions at the surface of LDHs are influenced by the composition of CO₂. Water vapor can enhance CO₂ adsorption efficiency by promoting the formation of bicarbonates through reactions with metal hydroxides in LDHs under humid conditions (Cheah et al. 2022; Jerome et al. 2022).

The inclusion of carbon within MMOs in the nanocomposite effectively prevented particle aggregation and fusion throughout the cycling process, thereby maintaining its capacity to collect substances across several cycles. To use these sorbents in an industrial environment, it is critical to evaluate their efficacy and resilience when exposed to elevated levels of CO₂ in flue gas. The nanocomposite

Fig. 19 Changes in LDO sorption performance from wet to dry CO₂ environments (Ram Reddy et al. 2008)



demonstrated comparable cycling stability, with an 86% CO₂ capture rate, but saw a decrease in cycling capacity from 5.10 to 4.10 mmol/g. To summarize, the organic–inorganic nanocomposite has exceptional capabilities for absorbing and preserving CO₂ while remaining stable in both CO₂-rich and CO₂-lean (flue gas) conditions (Pelech et al. 2022). Finally, the feed gas composition can influence the regeneration and stability of LDH. Higher CO₂ concentrations can lead to more stable carbonate formations, potentially affecting the material thermal stability and its ability to be regenerated (Wang et al. 2022b; Fierro et al. 2023). Understanding the interplay between different gases and CO₂ adsorption in LDHs is crucial for optimizing their performance in capturing CO₂ from mixed gas streams.

Mechanism of CO₂ capture using LDH-based materials

LDHs are versatile and promising materials for CO₂ capture, attributed to their unique structural properties and functionality (Tang et al. 2022; Ge et al. 2023). These materials are characterized by their anion-exchange ability, high positive surface charge, and a distinctive structure composed of positively charged layers with intercalated anions located within their interlayer spaces. Such a configuration enables LDHs to engage in both physical and chemical adsorption of CO₂, leveraging their high surface charge and the capability to modify interlayer anions to optimize CO₂ capture processes (Marcu and Pavel 2022; Liu 2022; Tang et al. 2022; Fan et al. 2023; Ge et al. 2023; Sharma et al. 2023; Zaker et al. 2023).

The process of CO₂ capture utilizing LDH-based materials involves the initial adsorption of CO₂ molecules onto the LDH surface through Van der Waals forces, a process influenced by the LDH surface area, pore size, and structural

properties (Zhang et al. 2023b; Tao et al. 2023). Furthermore, the process involves the insertion of CO₂ molecules into the interlayer spaces of LDH, where the layers negative charge attracts positively charged CO₂ molecules via electrostatic forces, enhancing the capture efficiency (Palin et al. 2019; Sarmet et al. 2023). This is followed by the transformation of these molecules into carbonate or bicarbonate anions, a conversion enhanced by the presence of water (Wu et al. 2022a; Wang et al. 2022a; Zhao et al. 2023a; Zhang et al. 2023b; Tao et al. 2023). The transformation into easily stored, chemically stable carbonates on the LDH surface is primarily driven by hydroxyl groups, allowing CO₂ to react and form carbonates (Min et al. 2023; Ge et al. 2023; Kamada et al. 2023; Priyadarshini et al. 2023). These carbonates are then intercalated into the LDH structure, underscoring the critical role of the LDH metal cation composition in its CO₂ adsorption and conversion efficiency (Zhang et al. 2023b; Tao et al. 2023). These hydroxyl groups not only increase the surface reactivity of LDHs with CO₂ but also allow for the formation of hydrogen bonds, which further enhance CO₂ adsorption. Hydrogen bonding between the hydroxyl groups on LDHs and CO₂ molecules is pivotal for the stability and retention of CO₂ within the LDH structure, preventing its release back into the atmosphere (Manchala and Veldurthi 2023). The reactions involved in the process can be depicted by the following chemical equations:

1. The conversion of CO₂ via its interaction with hydroxide ions into bicarbonate ions:



2. The process of CO₂ reacting with bicarbonate ions to produce carbonate ions:

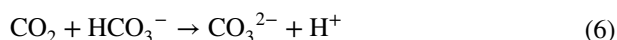
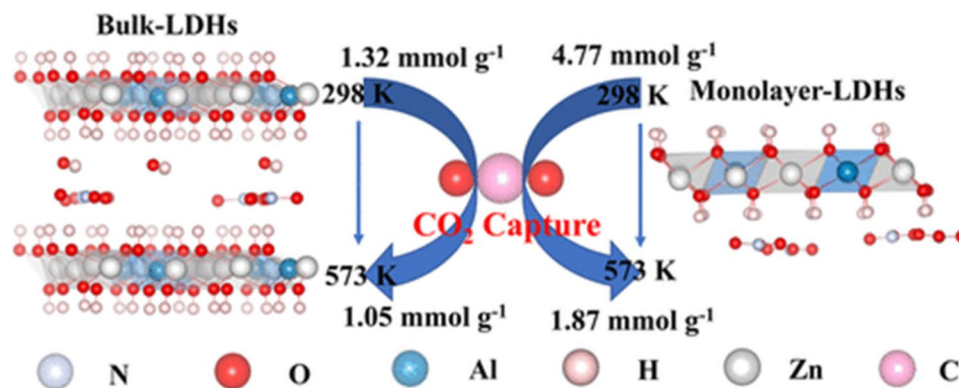


Fig. 20 CO₂ adsorption by bulk- and monolayer-LDHs (Zhang et al. 2020b)



Zhang et al. (2020b) investigation into CO₂ adsorption by bulk and monolayer-LDHs through density functional theory (DFT) highlighted differences in adsorption modes due to variations in charge distribution, density of states, and charge transfer characteristics (Zhang et al. 2020b). Bulk-LDHs predominantly utilize chemical adsorption, characterized by a higher adsorption energy (−1.01 eV) resulting from CO₂ interacting with the 5d and 6p electrons of LDHs, which enhances electron distribution near the Fermi level and creates new electronic states.

Conversely, monolayer-LDHs rely on van der Waals forces for physical adsorption, marked by minimal electron transfer. Thermodynamic analysis, focusing on entropy (ΔS), enthalpy (ΔH), and Gibbs-free energy (ΔG) changes, indicated that temperature plays a crucial role in CO₂ adsorption dynamics. Below 400 K, both bulk and monolayer LDHs show similar entropy changes, but as temperatures rise, bulk-LDHs exhibit a higher CO₂ adsorption capacity, whereas monolayer-LDHs, with their vast surface area and electronic activity, are more adept at capturing CO₂ at lower temperatures through physical adsorption, also indicating potential for CO₂ reduction catalysis (Zhang et al. 2020b). This analysis, supported by Fang et al. (2020) findings on CO₂ adsorption preference on LDH surfaces in a parallel configuration highlight the complex relationship between structural and electronic properties in dictating CO₂ adsorption efficiency and mechanisms on LDHs (Fang et al. 2020). Figure 20 illustrates the process of CO₂ adsorption by both bulk and monolayer forms of LDHs.

The CO₂ adsorption process on MMOs at moderate temperatures is a combination of physisorption and chemisorption (Santamaría et al. 2023). Initially, a chemisorbed layer forms, characterized by a strong bond; subsequently, layers are physisorbed, exhibiting weaker binding strengths (Cheah et al. 2022; Hitam and Aziz 2023). The extent of chemical interactions, limited at lower temperatures due to the requirement for higher activation energy, increases with temperature, enhancing chemisorption (Chávez-Rocha et al. 2023). Conversely, physical adsorption decreases with rising

temperatures, attributed to negative enthalpy (Lund et al. 2022). This interplay between physisorption and chemisorption at different temperatures is crucial in understanding the CO₂ adsorption behavior on LDH-based MMOs, highlighting the importance of temperature in influencing the adsorption process (Veerabhadrapa et al. 2021).

Calcination temperature and the presence and ratio of M²⁺ and M³⁺ cations significantly impact CO₂ adsorption mechanisms in LDH materials. Calcination exposes basic sites on the MMO surface, enhancing physisorption related to site density (Ai et al. 2023). The nucleophilicity of surface functional groups, influenced by basicity, affects chemisorption efficiency without directly increasing CO₂ capture (Okello et al. 2023). Isosteric heat, indicating the strength of CO₂ binding to the adsorbent, is crucial for assessing adsorption efficiency. Ideally, this binding energy ranges from 35 to 50 kJ/mol, a range typical for MMOs (Xu et al. 2023). Higher isosteric heat values necessitate high-temperature regeneration processes, similar to lithium zirconate or calcium oxide, due to energy penalties (Tan et al. 2023). Conversely, lower isosteric heat values result in reduced efficacy at low CO₂ partial pressures (Santamaría et al. 2023).

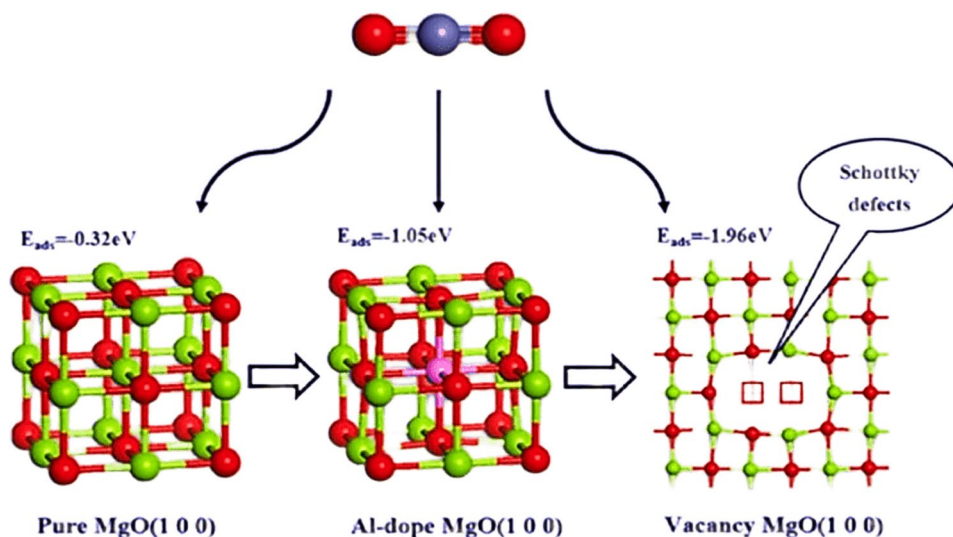
Another study delves into the effectiveness of MMO-based adsorbents, particularly those derived from LDHs, in capturing CO₂. It is well understood that LDHs by themselves are not sufficient for CO₂ capture unless they are thermally treated or modified structurally. The effectiveness of CO₂ adsorption and desorption on mixed metal oxides hinges on the basic site M–O*, which plays a pivotal role in the formation of both monodentate and bidentate carbonate ligands on the surface, using the Mg–O site as an illustrative example (Tan et al. 2024).



Research by Colonna et al. (2018) highlighted that among various LDH-derived MMOs (Zn–Al, Zn–Cu–Al, Cu–Al, Ni–Al), the Cu–Al–MMO showed superior CO₂ adsorption performance, with significant improvements under both

Fig. 21 The reactive species Mg–O formation mechanism (Lv et al. 2022)

Mechanism of CO₂ adsorption on point-defective MgO surfaces: First-principles study



pure CO₂ and atmospheric conditions. The study noted the simultaneous adsorption of CO₂ and water in all samples, with the adsorption ratio varying according to the specific MMO (Colonna et al. 2018). Qin et al. (2017b) found that the length of carbon chains in intercalated carboxylic acid anions within Mg–Al-LDHs affected CO₂ capture performance, observing an increase in CO₂ capture with carbon chain lengths greater than 10, peaking at a carbon chain length of 16. This enhancement is attributed to the transformation of LDH-C16 platelet structure into nanoparticles upon calcination, thus exposing more basic sites for CO₂ adsorption (Qin et al. 2017b).

Further investigations by Lv et al. (2022) expanded on these findings by exploring the impact of Al doping on CO₂ adsorption. They discovered that Al doping facilitates the orbital hybridization of the O atom in CO₂ with surface Mg atoms, resulting in the formation of a new Mg–O chemical bond. Additionally, the presence of vacancies was shown to encourage orbital hybridization between C and O atoms in CO₂ with surface O and Mg atoms, leading to the formation of C–O and Mg–O bonds, respectively. Notably, the study found that the Mg–O bond created through vacancies is stronger than that formed through Al doping, as evidenced by crystal orbital Hamiltonian populations analysis. This research underscores the intricate interplay between calcination temperature, Al doping, and the presence of vacancies in optimizing the CO₂ adsorption capacity of mixed metal oxides (Lv et al. 2022). Figure 21 illustrates the process by which the reactive species Mg–O is formed.

The exploration of CO₂ adsorption mechanisms by LDOs is hindered by the limitations of traditional experimental methods, particularly in elucidating processes at

the molecular scale. To bridge this gap, molecular dynamics (MD) simulations emerge as a potent tool, offering detailed insights into molecular-level interaction (Lv et al. 2022). Previous research utilizing MD simulations has underscored the essential role of water in augmenting CO₂ capture capabilities of Mg–Al–Cl LDH under specific conditions (Khorshidi et al. 2018). Further investigations revealed the structural transformations of Mg–Al–CO₃ LDH during heat treatment (Gao et al. 2018) and examined CO₂ adsorption behaviors on LDO, highlighting a comparison between the adsorption capacities of crystalline versus amorphous LDOs formed post-calcination. It was discovered that amorphous LDOs generally possess superior physical adsorption capacities (Khalkhali et al. 2020). However, despite these advancements, MD simulations encounter challenges in detailing the chemisorption mechanism of CO₂ adsorption by LDO at the electronic level, pointing to a need for complementary methods to fully decipher this complex process (Lv et al. 2022).

Gao et al. (2013) demonstrated that the calcination temperature is key to forming active Mg–O species for effective CO₂ adsorption, with high temperatures leading to MgAl₂O₄ spinel oxide and low temperatures preserving the hydroxide phase. They proposed two mechanisms for generating active sites at optimal temperatures: substituting Mg²⁺ with Al³⁺ in the periclase MgO lattice and shifting Al³⁺ from octahedral to tetrahedral sites, thus creating vacancies. This insight into the temperature-dependent formation of active sites and the strategic elemental substitution highlights the nuanced approach required to optimize LDH-based materials for CO₂ capture (Gao et al. 2013). When LDHs undergo calcination to form LDOs, the Mg–O and Al–O groups transform.

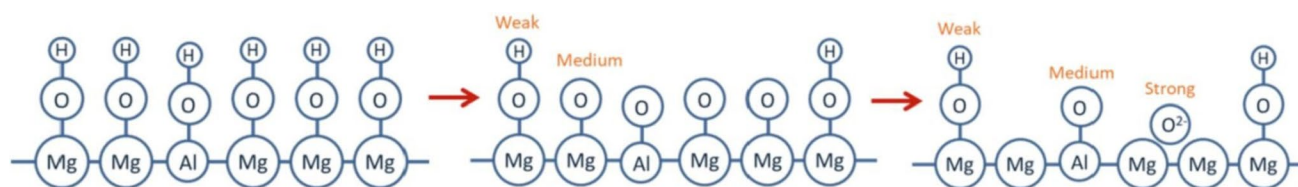


Fig. 22 Possible mechanisms for the creation of simple sites during LDHs thermal decomposition (Wang et al. 2022a)

Mg–O bonds are more prone to breakage than Al–O bonds, resulting in the creation of coordinated unsaturated ion O^{2-} , forming strong basic sites in LDO materials. Upon calcination at 450 °C, LDO materials exhibit a range of basic sites, including weak, medium, and strong categories, as depicted in a Fig. 22. This variation in basic site strength contributes significantly to the diverse catalytic and adsorption properties of LDO materials, enhancing their versatility in various applications (Wang et al. 2022a; Manríquez-Ramírez et al. 2022; Ibrahimova 2022; Kameliya et al. 2023).

Matsuda et al. (2023) detailed the multi-step transformation during thermal decomposition of Mg–Al LDH, involving water release, dehydroxylation, and collapse of the layered structure. LDH structures undergo thermal decomposition, and the reduction in interlayer spacing due to the removal of interlayer water is a characteristic of the amorphous phase (Matsuda et al. 2023). Upon reaching 400 °C, the LDH transforms into an amorphous mixed-phase oxide named $MgAlO_x$ (Cheah et al. 2022). Subsequent heating to 800 °C leads to a partial conversion, resulting in the formation of $MgAl_2O_4$ spinel at 950 °C (Zeng et al. 2022). This detailed understanding of the structural changes during thermal treatment provides insights into the transformation of LDH into metal oxides (Wiyantoko et al. 2022).

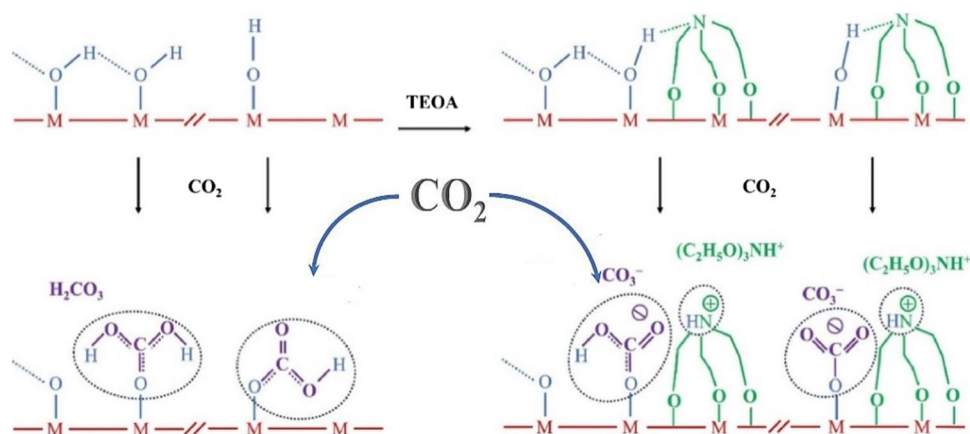
In situ diffuse reflectance infrared Fourier transform spectroscopy (DRIFTS) was employed to explore the CO_2 adsorption mechanisms on various adsorbent samples, including Santa Barbara Amorphous-15 (highly stable mesoporous silica) SBA-15, 40LDH/SBA-15, 40LDH-2PEI/SBA-15, MA, and 30LDH/MA (Wu et al. 2022a, b). Notably, the 40LDH-modified SBA-15 samples displayed intensified bands for carbonate or bicarbonate species, suggesting the formation of interlayer carbonate anions within the LDH structure (Wu et al. 2022b). Further DRIFTS analysis after 120 min of CO_2 adsorption highlighted the presence of both monodentate and bidentate carbonates, along with a smaller quantity of bicarbonate, with adsorption bands intensifying initially but slowing over time, aligning with thermogravimetric (TG) adsorption process observations. Despite similar positions of the main adsorption bands across the MA and 30LDH/MA samples, differences in intensity were noted, indicating that LDH loading altered the strength but

not the nature of the adsorption sites. This comprehensive analysis underscores that CO_2 adsorption on these samples results predominantly in the formation of carbonate species, showcasing the alteration in adsorption site strength upon LDH incorporation into the composite materials without changing the nature of the adsorption site (Wu et al. 2022a).

Further DRIFTS measurements on triethanolamine (TEOA-LDO) identified peaks indicative of HCO_3^{3-} and carbonate (CO_3^{2-}) species, alongside evidence of NH^+ formation, suggesting that the adsorbed CO_2 undergoes a transformation predominantly into HCO_3^{3-} and CO_3^{2-} on TEOA-LDO surfaces. This transformation is facilitated by an alkali-catalyzed hydration process, where protons from H_2CO_3 and HCO_3^{3-} are transferred to the tertiary amines in TEOA, leading to the formation of more stable anionic forms. Moreover, the DRIFTS analysis showed that in the absence of water, the intensity of NH^+ peaks decrease over time, while the $-OH$ group signal intensifies, indicating ongoing reactions between surface $-OH$ groups and CO_2 to form HCO_3^{3-} , which is further stabilized through proton donation to tertiary amines, resulting in NH^+ and CO_3^{2-} formation. This study illuminates the complex interaction mechanisms between CO_2 and the adsorbent surfaces, revealing that on Ni-Co-Al-LDO surfaces, H_2CO_3 and HCO_3^{3-} are the primary reaction products, whereas on TEOA-LDO surfaces, the process yields predominantly HCO_3^{3-} and CO_3^{2-} as illustrated in Fig. 23. It underscores the significant role of tertiary amines in stabilizing adsorbed CO_2 by converting it into more stable anionic species through a proton transfer process, with each CO_2 molecule being stabilized by a single amine group (Zhao et al. 2023a).

However, it is important to note that while the chemistry is well understood, the practical application of this process on a large scale still presents many challenges that require further research and development (Fan et al. 2022). These include the need for efficient regeneration of the LDHs after CO_2 capture, the stability of the LDHs under operating conditions, and the overall cost and energy efficiency of the process (Jerome et al. 2022; Cardinale et al. 2023a). Addressing these challenges will be crucial for the successful large-scale implementation of LDH-based CO_2 capture and utilization processes, emphasizing the need for continued advancements in this field.

Fig. 23 Illustration of CO₂ adsorption mechanism on Ni-Co-Al-LDO and TEOA-LDO surfaces (Zhao et al. 2023a)



Stability of CO₂ captured in LDH-based materials

LDHs have garnered significant attention as highly promising materials for CO₂ capture, owing to their distinct structural features that enable efficient sequestration. The structure of LDHs is characterized by positively charged metal hydroxide layers, intercalated with various anions and water molecules. CO₂ is typically captured in the form of carbonate anions within the interlayer spaces, where robust electrostatic interactions between the metal hydroxide layers and the carbonate ions play a crucial role in stabilizing the sequestered CO₂ (Kameliya et al. 2023).

Under ambient conditions, the stability of CO₂ within LDHs is well-maintained. However, thermal perturbations, particularly at elevated temperatures, can prompt the release of CO₂. This release is frequently associated with the decomposition of carbonate anions or structural phase transitions within the LDH material that disrupt the interlayer environment. The inherent chemical stability of LDHs is generally strong, offering substantial resistance to external environmental factors. Nonetheless, the stability of the captured CO₂ can be compromised by the presence of water vapor or acidic gases, such as sulfur dioxide (SO₂), which can initiate competitive adsorption processes that displace the CO₂ from the interlayer spaces (Zhu et al. 2021b). A notable characteristic of LDHs is the reversibility of the CO₂ capture process, which facilitates the regeneration of the material after CO₂ release. This reversible capture is particularly advantageous for applications requiring multiple cycles of CO₂ capture and release. However, while this reversibility enhances the material practicality in dynamic applications, it poses potential challenges for long-term CO₂ storage, as the ease of regeneration could inadvertently lead to the release of CO₂ under certain conditions (Kim et al. 2023).

LDHs stand out as promising candidates for CO₂ capture due to their unique structural attributes, which confer stability to the captured CO₂ across various conditions. The

ability of LDHs to undergo regeneration makes them especially suitable for cyclic CO₂ capture applications. However, for scenarios that demand sustained and permanent CO₂ retention, further research is necessary to address the challenges associated with the material reversible nature and to optimize its long-term performance (Marcu and Pavel 2022).

Regeneration of LDH-based materials

The regenerability of LDH-based materials after CO₂ adsorption is an important economic aspect. The ability to regenerate LDHs after CO₂ adsorption is critical for their practical application in cyclic carbon capture processes. Regeneration efficiency hinges on several factors, including the thermal stability of the LDH structure, the energy required to desorb CO₂, and the preservation of the LDH physical and chemical characteristics post-regeneration (Ge et al. 2023; Kim et al. 2023). The regeneration process involves removing the adsorbed CO₂ and restoring the LDHs to their original state, which can be achieved through various methods, including thermal treatment to release adsorbed CO₂ without structural damage (Bhatti et al. 2023); solvent washing with water, alcohol, or mild acids for gentle desorption (Lim et al. 2023a); vacuum application; or flushing with an inert gas like nitrogen or argon or steam to displace the adsorbed CO₂ (Kameda et al. 2021). Pressure swing adsorption (PSA) and electric swing adsorption (ESA) are critical technologies for regenerating layered double hydroxides (LDHs) used in CO₂ capture, each harnessing unique mechanisms to optimize efficiency and sustainability across various industrial applications (Fouladi et al. 2020). PSA operates on the principle of cyclical pressure changes to adsorb CO₂ at high pressures and then desorb it by significantly lowering the pressure, efficiently preparing the LDHs for subsequent cycles (Wiheeb et al. 2016). This method not only maintains the structural integrity and stability of LDHs under the stress of constant pressure fluctuations but

also finely tunes pressure settings to sync with the operational demands of the system, ensuring an optimal balance between operational efficiency and energy usage (Khajuria and Pistikopoulos 2011). On the other hand, ESA introduces a novel approach by using electrical energy to induce the desorption of CO₂, rapidly altering the physical or chemical properties of the adsorbent material. This method provides a swift and energy-efficient alternative to traditional thermal or chemical techniques, offering controlled and rapid desorption that enhances cyclic operation efficiency (Zhao et al. 2018). However, ESA also presents significant challenges, particularly in optimizing electrical parameters to maximize CO₂ desorption without compromising the adsorbent integrity, and in scaling the technology for large-scale operations, which requires substantial engineering solutions to overcome economic and technical barriers (Zhao et al. 2019). These methods are also applicable to the regeneration of carbonaceous adsorbents, with thermal regeneration being the most traditional and widely used (Salvador et al. 2015).

Various studies highlight critical factors that influence the regeneration efficiency of LDHs in CO₂ capture applications, emphasizing the importance of maintaining the material CO₂ capture capacity over multiple cycles (Garcia-Gallastegui et al. 2012a; Cardinale et al. 2023a). One key aspect is regeneration efficiency, where processes such as thermal treatment have been shown to restore the Mg–Al mixed-oxide CO₂ sorption to nearly its initial capacity after several cycles (Tang et al. 2018). The thermal treatment method is widely employed for regenerating LDHs post-CO₂ capture, involving heating the LDH-based adsorbent to specific temperatures to expel captured CO₂ and revert the adsorbent to its original state (Ram Reddy et al. 2006). This method prompts a structural evolution in LDHs, undergoing dehydration, dehydroxylation, and decarbonation stages, eventually transforming into an LDO structure, and further into a periclase-like (MgO) structure at higher temperatures (Jijoe et al. 2021; Kasneryk et al. 2023).

Equally important is preserving the structural integrity and composition of LDHs during regeneration, as any alterations could impact their CO₂ capture capabilities. Methods like calcination at specific temperatures have led to phase transformations in LDHs, enhancing their sorption capacities, while solvent washing presents another avenue for regeneration, with the choice of solvent playing a significant role in the process effectiveness (Cardinale et al. 2023a). These methodologies underscore the necessity of a balanced approach to regeneration that considers not only efficiency and capacity maintenance but also factors like scalability, cost, and environmental implications to ensure the practical application of LDHs in CO₂ capture.

A study by Wu et al. (2022a) investigated the impact of regeneration temperature on the CO₂ regeneration efficiency of a 30LDH/MA composite sample. The results showed that

lowering the regeneration temperature to 120 °C does not negatively affect the efficiency, suggesting a reduced temperature is preferable for this composite material. Recycling experiments were conducted over 11 cycles, revealing that the composite maintained a high regeneration efficiency of over 89%, outperforming the LDH sample (79%) and marginally below the MA (93%). These results highlight the excellent regeneration performance of the composite, highlighting its promising application in CO₂ capture technology. The composite ability to sustain high regeneration efficiency over multiple cycles, especially at lower temperatures, offers an effective and energy-efficient solution for CO₂ adsorption and recycling, contributing to the development of sustainable industrial processes (Gunathilake et al. 2016).

The effectiveness of CO₂ recovery using modified LDHs was evaluated, specifically UB-MEA5 and UH-MEA5, developed through sonochemical methods. During the desorption phase conducted at 105 °C for 30 min under a nitrogen atmosphere, UH-MEA5 showed superior initial CO₂ adsorption but had a lower recovery rate than UB-MEA5. Both materials exhibited a cyclic adsorption capacity where LDH-MEA5 maintained more than 90% of its initial capacity up to the 11th cycle but then declined to less than 60%. This reduction was attributed to a secondary reaction forming stable polyurea on the adsorbent surface, which was more pronounced in LDH-MEA5 compared to UB-MEA5 and UH-MEA5 (Cui et al. 2023).

The research also examined the performance of amine-grafted adsorbents, namely LDH-MEA5, UB-MEA5, and UH-MEA5, with varying adsorption periods. The secondary reaction led to the creation of persistent polyurea compounds, which could potentially cause degradation of LDH-MEA5 when subjected to several TSA cycles, especially when adsorption periods increase. Conversely, UH-MEA5 showed a significantly lower initial cyclic adsorption capacity, which it maintained for 30 TSA cycles. The CO₂-TPD profiles substantiated the results, demonstrating that UH-MEA5 exhibited a slightly higher CO₂ desorbed capacity compared to LDH-MEA5. Furthermore, the overall desorption was more prominent in the adsorbents subjected to ultrasonic irradiation (Rocha et al. 2019).

Another study focused on comparing the adsorption performance of sonochemically generated LDHs to that of conventionally synthesized LDHs, assessing their efficacy through kinetic models. The experimental outcomes aligned closely with the pseudo-second-order kinetic model for non-functionalized adsorbents, while data from UH-MEA5, post-amine extraction, conformed to the pseudo-first-order model, underscoring the straightforward recovery of CO₂ due to infrequent chemisorption (Della Moretta and Craig 2022). Sonochemically synthesized UB-MEA5 and UH-MEA5 exhibited superior gas adsorption capabilities, likely attributed to their increased pore volumes, a consequence

of the sonochemical processes used in their creation. This enhancement suggests that sonochemical methods hold considerable promise in improving CO₂ capture efficiency (Chávez-Rocha et al. 2023; Priyadarshini et al. 2023).

In conclusion, the exploration of the mechanism of CO₂ capture using LDH-based materials, along with the optimization of adsorbent regeneration processes, represents a vital research endeavor with significant environmental and economic implications. Future advancements in this field will require a multidisciplinary approach, combining material science, chemistry, and engineering, to realize the full potential of LDHs in combating global CO₂ emissions.

Future outlooks

The present study reviews the significant advancements in CO₂ capture using LDH-based materials and highlights the need for further research to overcome existing challenges and limitations. This proposed future research encompasses several key areas crucial for optimizing the performance of LDH-based materials in CO₂ capture applications. The key points are summarized below.

1. A deeper understanding of the CO₂ capture mechanisms is imperative. Although mixed metal oxides derived from LDHs have shown promising results in terms of CO₂ adsorption, stability, and regenerability, a comprehensive understanding of the underlying processes remains limited. Future research should prioritize both experimental and theoretical investigations to elucidate the fundamental mechanisms of CO₂ capture. This involves exploring the kinetics and thermodynamics of CO₂ adsorption, understanding the influence of metal cations, and assessing how material structure affects performance.
2. Addressing the impact of other gaseous substances on CO₂ capture is crucial. While much of the current research has focused on pure CO₂, real-world applications involve complex gas mixtures including CO, NO_x, SO_x, and water vapor. Future studies should integrate these variables to better simulate industrial flue gas conditions. Developing accurate molecular models and force fields to represent these conditions will enhance the testing and optimization of LDH-based materials, ensuring they perform effectively in the presence of interfering gases.
3. Enhancing knowledge on co-adsorption and selectivity is essential. The current understanding of CO₂ co-adsorption with other gases using LDH-based materials is limited. Systematic investigations are needed to identify parameters affecting co-adsorption and to design materials with improved selectivity for CO₂. Future research should focus on optimizing operating conditions to maximize the effectiveness of LDH-based materials in capturing CO₂ from complex gas mixtures.
4. In comparison to monoethanolamine (MEA), a widely used CO₂ capture agent, LDH-based materials show potential but currently fall short in terms of performance. MEA is known for its efficiency but suffers from high energy consumption and degradation. Research should aim to improve the cyclic efficiency of LDH-based materials, potentially through the incorporation of functional groups or organic molecules that enhance CO₂ adsorption. Advancements in the formulation of material composites and synthesis techniques are necessary to achieve superior performance and reduce production costs.
5. Optimizing the design of LDH-based materials for industrial applications is another crucial area for future research. This involves addressing various considerations such as energy consumption during synthesis, production yield, material durability, and the development of eco-friendly synthesis processes. Employing molecular dynamics simulations and computational tools (Gawish et al. 2022; Ganiyu et al. 2023; Ismail et al. 2023, 2024a, b; Al-Amrani and Onaizi 2024) can assist in predicting optimal synthesis routes and expected performance. Emphasis should be placed on designing materials that are not only effective but also cost-efficient and durable for commercial use.
6. With the enforcement of more stringent regulations, effective waste treatment (Al-Qadri et al. 2022; Al-Jamimi et al. 2023; Ibrahim et al. 2024; Al-Nowaiser et al. 2024; Alkoshab et al. 2024; Aziz et al. 2024) and waste utilization (Ahmed et al. 2022a, b; Onaizi 2023a, 2024a; Iddrisu et al. 2024), in addition to the utilization of sustainable materials (Onaizi et al. 2007, 2009, 2010, 2014a, b, 2016a, b; Onaizi 2017, 2018, 2021, 2023b, 2024b; Qasem et al. 2018; Alshabib and Onaizi 2019; Lateef et al. 2019; Ahmed et al. 2021; Drmoseh et al. 2022; Al-sakkaf and Onaizi 2023), are gaining a growing research interest. Thus, effective treatment approaches of the waste generated during the synthesis of LDH-based materials should be developed. Additionally, developing green synthesis routes and utilizing renewable energy sources in the production of LDHs-based materials should be considered in future research. Furthermore, utilizing waste materials for the production of LDH-based materials presents a promising opportunity for both environmental sustainability and economic benefit. This approach can aid in waste management while converting waste into valuable CO₂ capture materials. Future research should explore suitable waste streams and develop efficient processes for converting them into high-quality LDHs, thereby enhancing the economic and environmental viability of CO₂ capture technologies.

7. Transitioning from batch to continuous production processes is also recommended to improve cost-effectiveness and scalability. Continuous production methods can leverage economies of scale and reduce production costs. Additionally, innovative and cost-effective regeneration processes for spent LDH-based materials should be developed to enhance the overall efficiency and feasibility of CO₂ capture systems.
8. Expanding studies on the reusability and practicality of LDH-based materials is crucial. Current research often focuses on a limited number of adsorption-regeneration cycles. Future investigations should extend these studies to evaluate material performance over a larger number of cycles and under real-world conditions. Additionally, exploring applications for CO₂-laden regeneration gases could maximize the overall efficiency and economic benefits of the CO₂ capture process.

Addressing the above-mentioned research gaps could significantly advance the development and application of LDH-based materials for CO₂ capture. By focusing on these aspects, the scientific community can overcome current limitations and harness the full potential of LDH-based materials in tackling the global challenge of CO₂ emissions.

Conclusions

It is worth noting that LDHs present a unique and versatile platform for CO₂ capture applications characterized by their layered structure and anion exchange capabilities. LDHs possess favorable properties for CO₂ capture, including high surface area, tunable chemical composition, and excellent stability under relevant operating conditions. Additionally, LDHs exhibit strong CO₂ adsorption capacities, thermal stability, ease of preparation, tunable properties, and high selectivity over other gases, making them attractive candidates for carbon capture and storage applications. The chemical composition, synthesis conditions, and methods of the LDHs significantly influence their CO₂ capture ability. The most effective synthesis techniques for LDHs aimed at CO₂ adsorption are co-precipitation and urea hydrolysis. Co-precipitation methods are noted for producing LDHs with excellent yield, purity, and CO₂ adsorption efficiency. Although LDHs created through urea hydrolysis demonstrate enhanced cyclic stability, they tend to have lower yields and purity because of amorphous by-products. Several techniques have been investigated to modify LDHs for CO₂ adsorption. These techniques include alkali impregnation, calcination, and the creation of composites containing LDHs. The goal of these modifications is to improve the CO₂ adsorption capacity, selectivity, and stability of LDH-based materials. This research

contributes to the development of CO₂ capture technologies, which are important for mitigating climate change. Alkali impregnation involves the incorporation of alkali metals into the LDH structure, enhancing its CO₂ adsorption capacity by introducing additional active sites and improving the material basicity. Calcination, on the other hand, involves the thermal treatment of LDHs to generate MMOs, which exhibit enhanced textural properties and surface areas, leading to improved CO₂ adsorption performance. Moreover, the development of composites containing LDHs combines the advantages of both materials, such as high surface area and tunable pore structures, to achieve synergistic effects in CO₂ capture. Composites, especially those made of GO, choosing an M²⁺/M³⁺ ratio above 4, choosing to improve the LDH exfoliation process, and choosing magnesium aluminum (Mg–Al) samples are all things that could be looked into further. Novel approaches, including the use of amino-functionalized LDHs, have demonstrated improved CO₂ adsorption performance. Additionally, the facile synthesis of LDH nanosheets grafted with aminosilane has shown good CO₂ adsorption capacity and stability. Ongoing research continues to explore novel modification strategies to further optimize the performance of LDHs for CO₂ adsorption applications. Furthermore, the understanding of the adsorption mechanisms involved in LDHs has been illustrated, providing insights into their potential for practical implementation in mitigating CO₂ emissions. Overall, research findings underscore the potential of LDHs as effective materials for CO₂ adsorption, paving the way for their utilization in addressing environmental challenges associated with greenhouse gas emissions.

Author contribution Mehrab A. Hassan: investigation, analysis, visualization, validation, formal analysis, writing (original draft).

Sheikha F. Wahdain: investigation, analysis, visualization, validation, formal analysis, writing (original draft).

Sagheer A. Onaizi: visualization, validation, supervision, resources, project administration, investigation, funding acquisition, conceptualization, writing (review and editing).

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Declarations

Ethical approval and consent to participate This study does not include any human or animal subjects.

Consent for publication Not applicable.

Competing interests The authors declare no competing interests.

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